

An Endpoint Gradient Bound for the Centered Disk Maximal Operator

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September 26, 2026

Abstract

We prove the endpoint gradient bound $\|\nabla Mf\|_1 \leq C\|\nabla f\|_1$ for every real-valued $f \in W^{1,1}(\mathbb{R}^2)$, where $Mf(x)$ is the supremum of the averages of $|f|$ over disks centered at x , and C is an absolute constant. The maximal function belongs to $W_{\text{loc}}^{1,1}(\mathbb{R}^2)$ and has a globally integrable weak gradient. This gives a positive resolution of the planar centered-disk case of the endpoint question of Hajlasz and Onninen.

1 Introduction

For $f \in L_{\text{loc}}^1(\mathbb{R}^2)$, the centered disk maximal function is

$$Mf(x) = \sup_{r>0} \frac{1}{\pi r^2} \int_{B(x,r)} |f(y)| \, dy, \quad B(x,r) = \{y \in \mathbb{R}^2 : |y-x| < r\}.$$

Every averaging disk is centered at the point of evaluation, and all positive radii are allowed. Kinnunen proved that this operator is bounded on $W^{1,p}$ for $1 < p < \infty$ [10, Theorem 1.4]. The pointwise control of its gradient by $M(|\nabla f|)$ gives the corresponding norm bound through the strong L^p maximal inequality. At $p = 1$ that last step fails. Hajlasz and Onninen asked whether the gradient estimate itself nevertheless survives [8, Question 1]. We give a positive resolution of the planar centered-disk case of their endpoint question.

Theorem 1.1. *There is an absolute constant $C < \infty$ such that, for every real-valued $f \in W^{1,1}(\mathbb{R}^2)$, the function Mf is locally integrable, belongs to $W_{\text{loc}}^{1,1}(\mathbb{R}^2)$, and satisfies*

$$\int_{\mathbb{R}^2} |\nabla Mf| \, dx \leq C \int_{\mathbb{R}^2} |\nabla f| \, dx. \quad (1.1)$$

In particular, its distributional derivatives are represented by globally integrable functions.

Here $W^{1,1}(\mathbb{R}^2)$ is the inhomogeneous Sobolev space: the input and its weak first derivatives are globally integrable. Vector norms are Euclidean. No radially or compact-support condition is imposed on the input. The maximal function itself need not be globally integrable, even for a nonzero nonnegative smooth function of compact support. The constant in (1.1) is finite and absolute; its optimal value is not determined. The argument below concerns disks in the plane.

1.1 The signed estimate behind the theorem

The intermediate estimate retains the sign of the input. For real $g \in C_c^\infty(\mathbb{R}^2)$, put

$$A_t g(x) = \frac{1}{\pi t^2} \int_{B(x,t)} g(y) \, dy, \quad S_{a,b} g(x) = \sup_{a \leq t \leq b} A_t g(x).$$

For fixed $0 < a \leq b < \infty$, this envelope is compactly supported and Lipschitz. The issue is to bound its gradient uniformly as the range of radii grows.

Proposition 1.2 (Signed finite-band estimate). *There is an absolute constant C such that, for every real $g \in C_c^\infty(\mathbb{R}^2)$ and $0 < a \leq b$ with b/a a power of two, including one,*

$$\int_{\mathbb{R}^2} |\nabla S_{a,b} g| \, dx \leq C \int_{\mathbb{R}^2} |\nabla g| \, dx. \quad (1.2)$$

The constant is independent of a , b , and the number of dyadic radius scales between them.

The proof of Proposition 1.2 is given in Section 5, after the local and multiscale estimates described below. A single radius, $a = b$, is covered directly by convolution. The signed form of Proposition 1.2 has a specific purpose in the final regularity argument. After extension to Sobolev inputs, one applies it to $g_Q = \chi_Q(g - c_Q)$, where c_Q is a local mean and χ_Q equals one on the small disks being tested. On those disks, $A_t g_Q = A_t g - c_Q$, so the corresponding envelopes have the same gradient. Poincaré's inequality then bounds $\|\nabla g_Q\|_1$ by the gradient of g on a slightly larger square. The function g_Q can take both signs. Signed averages also occur in Weigt's dyadic and uncentered cube estimates [19, 20]; their role here is to permit this local subtraction.

1.2 Earlier endpoint results

In one dimension, Tanaka proved an L^1 derivative bound for the uncentered maximal operator [17, Theorem 1]. Aldaz and Pérez Lázaro extended the result to inputs of bounded variation, established absolute continuity of the maximal function, and obtained the sharp variation constant 1 [2, Theorem 2.5]. Kurka proved a variation bound with an absolute constant for the centered operator on functions of bounded variation, together with local Sobolev regularity and an integrable derivative for Sobolev inputs [11, Theorem 1.2 and Corollary 1.4]. The ordering of intervals used in these arguments does not directly organize a family of centered disks.

In higher dimensions, Aldaz and Pérez Lázaro proved regularity for block-decreasing inputs, invariant under coordinate sign changes and nonincreasing in each positive coordinate, and uncentered maximal operators over balls of unconditional norms; their weak-gradient conclusion applies, in particular, to block-decreasing $W^{1,1}$ input and Euclidean balls [1, Theorem 11]. Luiro proved the L^1 gradient estimate for radial Sobolev input and the uncentered ball operator [13, Theorem 3.12]. Weigt established variation bounds for the dyadic maximal operator [19] and for the uncentered maximal operator over axis-parallel cubes [20, Theorem 1.1]. Their geometry and indexing differ from centered disks. Moreover, a variation estimate controls a derivative measure, which may still have a singular part. For fractional orders $0 < \beta < 1$, Beltran and Madrid established centered endpoint gradient bounds for radial $W^{1,1}$ input [3, Theorem 1.2]. Weigt subsequently proved endpoint gradient estimates for both the centered and uncentered operators [18, Theorem 1.1]. The positive order restricts overlap among optimizing balls in that proof; the mechanism degenerates at order zero.

Lahti and Weigt's conditional regularity theorem upgrades local bounded variation of the centered maximal function to local Sobolev regularity when the input is Sobolev [12, Theorem 1.2].

It assumes the requisite variation bound. We prove that bound and then remove the singular part directly using the signed localization above. Finally, boundedness and continuity of the gradient map are distinct questions. The one-dimensional continuity results of Carneiro, Madrid and Pierce for the uncentered operator [4], and of González-Riquelme for the centered operator [6], concern this further issue. Theorem 1.1 asserts the planar norm estimate.

1.3 How the finite-band estimate is proved

Write

$$u_t = A_t g, \quad h = (\nabla g) \, dx, \quad V = |h|(\mathbb{R}^2) = \int |\nabla g|. \quad (1.3)$$

Scaling reduces Proposition 1.2 to radii in $[2^{-m}, 1]$. The proof has one geometric input and two separate multiscale constructions. The first construction organizes scalar cancellation; the second chooses directions and pays for their changes in both scale and space.

Disk moments and concentration in a shell. At an interior maximizing radius, the radial derivative of u_t vanishes. A rotational moment vanishes on every disk. In complex notation these identities give a zero first moment, which is preserved by suitable positive mixtures of nearby disks. Variations of optimizing balls also produce weighted-gradient identities in Luiro's work [13, Lemmas 2.2 and 2.4]; we derive the identities needed for centered signed averages directly.

Identify $\mathbb{R}^2$ with $\mathbb{C}$, let x be the disk center, and let z be a point of its thin boundary shell. The purpose of the moment identity is visible in the affine factor

$$W_z(y) = 1 - \frac{\overline{y - x}}{\overline{z - x}}.$$

This factor vanishes at z . Inserting it in a difference of balanced disk kernels therefore makes mass near z inexpensive. This leads to the geometric question: when is almost all relevant shell mass concentrated near one point? A sweep by nested disks tangent at that point answers it outside one exceptional set of centers, uniformly over every radius in the band. Crucially, it requires a bound on positive mass only in an inner disk; the error mass is controlled separately.

These observations yield a local directional estimate. On a square of side r , decompose h into positive measures with directions in one cone and localized errors. Errors smaller than the observation scale have zero total vector mass; errors at scale r are charged their full variation. An error of variation b supported at scale $s \leq r$ is charged $b(s/r)^\alpha$, with a fixed $\alpha = 1/100$. The resulting sum controls the integral over the square of the supremum, over locally maximizing radii, of the gradient component pointing against the cone. Companion estimates handle band endpoints and the value oscillation

$$v_k(x) = \sup_{0 < t, t' \leq 4r_k} |u_t(x) - u_{t'}(x)|, \quad r_k = 2^{-k}.$$

Those estimates include the endpoint arguments and the comparison of successively narrower kernels; shell concentration alone does not supply them.

Scalar cancellation and a common price for each cell. Let ϕ_q be the product of the two one-dimensional tents at a vertex q of the level- k dyadic grid: each factor equals one at its vertex coordinate and decreases linearly to zero one mesh width away. Set $h_q = \int \phi_q \, dh$, $a_q = |h_q|$. The tents form positive partitions of unity and refine positively onto finer grids. We express vectors in each of finitely many bases formed by the two boundary directions of a cone. For each scalar

coordinate, we start with the actual signed density at the finest level. Pair its positive and negative parts in equal masses, retain the constant-sign remainder, and repeat this operation on the weighted remainders at each parent. This bottom-up construction retains the support and birth level of every pair, including cancellation inside the finest tents. Its identity is an identity of measures, so there is no unrepresented terminal density.

The masses paired at successive levels form a telescoping sum. Charging each pair at coarser observation scales with the factor $(s/r)^\alpha$ gives a nonnegative price D_Q for every dyadic cell, with

$$\sum_Q D_Q \leq CV. \quad (1.4)$$

One price works for every direction in the fixed finite palette. The local estimates then cost D_Q together with appropriate nearby vector masses a_q at the observation scale. Multiscale oscillation budgets already play a role in Kurka's one-dimensional proof [11, Lemma 1.5]; the scalar measure decomposition and its common prices are proved here.

Vector particles and the cost of choosing directions. The remaining vector masses require a different construction. Vector refinement makes their total increase with the level, while keeping it at most V . We realize these masses by a finite positive measure on refinement paths, adding mass when needed and recording its birth level. This construction preserves the complete path, including levels before birth, so that it can compare choices of direction made from the first scale onward. These vector particles differ from the scalar cancelling pairs: their role is to turn estimates along paths into sums of vector-mass charges over cells.

Two costs must be controlled. A change of direction starts a new run of radius scales and creates endpoint terms. Spatial variation of the choice of direction creates derivatives of its weights when we integrate by parts. Section 4 constructs a finite-state law at each point x and bounds sums of both costs, weighted by these vector masses, by CV , with the spatial suprema needed for the cell estimates. Vector-additivity defects control changes along particles, and comparison at nearby points transfers that control to the spatially varying law.

Assembly and cancellation of differentiated rows. After the common prices and the direction costs have been bounded, we assemble the radius bands. A run is a maximal consecutive interval of scales carrying the same direction. For each history, apply the local directional estimate to the supremum over the radii of every run, then average over histories. The common prices and the bounds for direction changes pay for the unfavorable components and the endpoints of runs.

For each run, subtract the average at its starting scale from its envelope; the endpoint estimates pay for the gradients of these baselines. The vector quantity to be integrated by parts is the sum of the resulting envelopes multiplied by their run directions. Its history weight depends on x . If $\pi_k(x; n, n')$ is the probability of passing from direction n to direction n' at step k , normalization of each row gives

$$\sum_{n'} \nabla_x \pi_k(x; n, n') = 0.$$

Before taking absolute values, we can therefore subtract the analogous vector sum for the preceding history, ending its last run at level $k - 1$. That sum is independent of the next direction. The difference contains only extensions of the last run and new runs, and is bounded by $C \sum_{l \geq k} v_l$. The local value estimate contributes the scale r_l , while a differentiated transition row has scale $1/r_k$. Their ratio r_l/r_k is exactly the geometric weight controlled by the direction-cost estimate. Together with the common-price budget, these bounds give (1.2).

1.4 From the finite-band estimate to a weak gradient

Approximation extends Proposition 1.2 to signed Sobolev inputs. Applying it to $g = |f|$ and increasing the radius bands first gives a finite vector Radon measure representing the derivative of Mf . To prove that this measure has no singular part, fix a compact null set and a positive radius threshold. The envelope over larger radii is Lipschitz, so its contribution vanishes as the neighborhood of the null set shrinks. For small radii, the signed cutoff and local mean subtraction described above reduce the cost to the integral of $|\nabla g|$ within a fixed multiple of the threshold distance. After shrinking the neighborhood, let the radius threshold tend to zero. This gradient integral then vanishes as well. Inner regularity gives absolute continuity of the derivative measure and proves Theorem 1.1.

Lower-radius Lipschitz truncations appear in Hajlasz and Malý [7, Lemma 8], and local/nonlocal splitting with subtraction of an input mean appears in Saari [14, Section 3.1]. The signed estimate above supplies the gradient-only localization used here.

Section 2 proves disk cancellation and the local estimates. Section 3 gives scalar refinement and common prices; Section 4 constructs the vector particles and spatial label law. Section 5 assembles the finite bands, and Section 6 passes to Sobolev input and removes the singular part. An analytic appendix records the Sobolev calculus, square localization, and vector-measure facts in the forms used in the final step.

1.5 Conventions

A constant denoted by C may increase between occurrences. All parameters, including cone apertures, finite direction nets and geometric neighborhood sizes, are fixed absolute choices. In a local lemma the explicitly listed cone and support constants are allowed dependencies. No constant depends on the number of radius scales.

For a vector measure ν , $|\nu|$ denotes its total variation. For a real number a , $a_+ = \max(a, 0)$. A supremum of nonnegative quantities over an empty set is zero. When ∇u_t is written, the radius is held fixed. We use the classical strong L^p maximal inequality for $p > 1$ [10, (1.2)]; a general account is given in [16, Chapter I]. The analytic appendix records the remaining standard inputs in the forms used here.

2 Cancellation at disks and three local bounds

Throughout this section, $g \in C^\infty(\mathbb{R}^2)$ is real valued, $h = \nabla g dx$, and $u_t = A_t g$. The averages are signed. No compact-support assumption on g is needed for the local estimates. Our goal is to control three quantities on a unit square: unfavorable directional gradients at locally maximizing radii, gradients at band maxima and endpoints, and differences of averages. The common input is a decomposition of the gradient into positive directions and error pieces, with small support and zero total mass below size one. The discount on the latter errors will make the scale sum in Section 3 finite. We fix

$$\alpha = \frac{1}{100}. \quad (2.1)$$

An equality of measures on an open set means equality of their restrictions there. The component measures themselves will always remain defined on the whole plane.

Lemma 2.1 (Local estimates). *Let Q be a square of side length one and let $\Omega = \{y : \text{dist}(y, Q) < 10\}$. Suppose that, on Ω ,*

$$h = \sum_i v_i \sigma_i + \beta, \quad \beta = \sum_d \beta_d. \quad (2.2)$$

The sums are finite. The measures in the sums are finite, defined on $\mathbb{R}^2$, and absolutely continuous with respect to Lebesgue measure. Each σ_i is nonnegative and each v_i is a constant unit vector. For each d , write $m_d = |\beta_d|(\mathbb{R}^2)$ and assign a number $s_d \in (0, 1]$. If $s_d < 1$, assume that $\beta_d(\mathbb{R}^2) = 0$ and that β_d is supported in a ball of radius at most $C_s s_d$, where C_s is fixed. Set

$$S = \sum_i \sigma_i, \quad A = \sum_d s_d^\alpha m_d.$$

For $x \in Q$, define

$$\begin{aligned} \mathcal{I}(x) &= \{r \in [1, 2] : u_r(x) \geq u_t(x) \text{ for every } t > 0 \text{ with } |t - r| \leq 1/4\}, \\ \mathcal{J}(x) &= \{r \in [1, 2] : u_r(x) = \max_{1 \leq t \leq 2} u_t(x)\} \cup \{1, 2\}. \end{aligned}$$

Then the following estimates hold.

(i) Unfavorable directional gradients. If e is a unit vector and $e \cdot v_i \geq \kappa > 0$ for all i , then

$$\int_Q \sup_{r \in \mathcal{I}(x)} (-e \cdot \nabla u_r(x))_+ dx \leq C A. \quad (2.3)$$

(ii) Band maxima and forced endpoints. Without a condition on the directions v_i ,

$$\int_Q \sup_{r \in \mathcal{J}(x)} |\nabla u_r(x)| dx \leq C(A + S(\mathbb{R}^2)). \quad (2.4)$$

(iii) Differences of nearby averages. We also have

$$\int_Q \sup_{0 < t, t' \leq 4} |u_t(x) - u_{t'}(x)| dx \leq C(A + S(\mathbb{R}^2)). \quad (2.5)$$

Here C depends only on C_s , and in the first estimate also on κ . A supremum of nonnegative quantities over an empty set is zero.

All the quantities in this statement are measurable. To see this, extend the displayed definitions of $\mathcal{I}(x)$ and $\mathcal{J}(x)$ to all $x \in \mathbb{R}^2$. Their graphs are closed in $\mathbb{R}^2 \times [1, 2]$, by joint continuity of $u_t(x)$ and continuity of a maximum over a compact interval. For either of the first two suprema, its superlevel set at a positive closed threshold is the projection of a closed set with the radius variable in a compact interval, and is therefore closed. Restricting these Borel functions to Q proves the claim also for half-open squares. The last supremum can be taken over positive rational $t, t' \leq 4$, by continuity in the radius variables.

The first estimate discounts cancellation without charging the positive mass S : its cone contribution is favorable at every tested radius. Comparisons of kernels will still require a bound on its weighted mass. The second estimate pays for S and also covers the two forced radii, so it can be used when a radius interval ends. The third controls values rather than gradients; its factor under rescaling will pay for derivatives of spatial weights. Section 3 gives those scaled applications.

We prove the first two bounds by comparing successively narrower disk kernels. Before estimating their changes, we explain why a maximizing radius supplies a useful moment identity and why concentration in a small cap makes that identity effective.

2.1 Disk moments and an affine cancellation

We seek cancellation in a thin shell where two nearby disks differ. Estimating by the entire gradient mass on that shell would ignore the maximizing property of the radius. At an interior maximizing radius, a radial derivative vanishes; a rotational moment vanishes for every disk. We combine these identities using complex notation, then preserve them while smoothing the disk boundary.

Variations of optimizing balls also give weighted-gradient cancellation identities in Luiro's work on the uncentered maximal operator [13, Lemmas 2.2 and 2.4]. There the balls optimize averages of $|f|$. We derive the identities needed here directly for centered signed averages.

We identify the plane with the complex numbers, using the usual positive orientation. A vector measure is then a complex measure, and $\overline{y-x} dh(y)$ denotes complex multiplication. For every disk,

$$\operatorname{Im} \int_{B(x,t)} \overline{y-x} dh(y) = 0, \quad \partial_t u_t(x) = \frac{1}{\pi t^3} \operatorname{Re} \int_{B(x,t)} \overline{y-x} dh(y). \quad (2.6)$$

The first identity follows by integrating the divergence of $g(y)(y-x)^\perp$: the vector field $(y-x)^\perp$ is divergence free and tangent to the boundary circle. For the second, differentiate $u_t(x) = \pi^{-1} \int_{B(0,1)} g(x+tw) dw$. The same change of variables gives

$$\nabla u_t(x) = \frac{h(B(x,t))}{\pi t^2}. \quad (2.7)$$

Lemma 2.2 (Balanced kernels). *Suppose that $r \in [1, 2]$ and $0 < \delta \leq 1/4$, and that*

$$u_r(x) \geq u_{r-\delta}(x), \quad u_r(x) \geq u_{r+\delta}(x).$$

There is a probability density k supported on $[r-\delta, r+\delta]$, bounded by C/δ , such that

$$K(y) = \int k(t) \mathbf{1}_{B(x,t)}(y) dt$$

satisfies

$$\int K(y) \overline{y-x} dh(y) = 0. \quad (2.8)$$

Moreover, $0 \leq K \leq 1$, $\operatorname{Lip}(K) \leq C/\delta$, $K = 1$ on $B(x, r-\delta)$, and $K = 0$ outside $B(x, r+\delta)$. The density may be chosen measurably in (x, r) .

Proof. Normalize t^{-3} separately on $[r-\delta, r]$ and $[r, r+\delta]$ to obtain probability densities k_- and k_+ . Let M_- and M_+ be the real moments in (2.6) averaged with these densities. Integration of that identity shows $M_- \geq 0$ and $M_+ \leq 0$. If $M_- - M_+ > 0$, take

$$k = ak_- + (1-a)k_+, \quad a = \frac{-M_+}{M_- - M_+};$$

if both moments are zero, take $k = k_-$. The imaginary moments are zero separately, so (2.8) follows. The normalizing factors are comparable to δ , because $3/4 \leq t \leq 9/4$. This proves the density bound and all the claims about K . The formula for a is Borel, with the stated convention at zero, which gives the last assertion. $\square$

The balanced kernels in Lemma 2.2 smooth the disk boundary without losing its zero moment. Consider two such kernels, possibly with different widths. Their difference is supported in the larger shell. The following elementary observation explains what we want to know about the mass on that shell.

Lemma 2.3 (Affine cancellation). *Let $x \in \mathbb{R}^2$, $r \in [1, 2]$, $0 < \delta \leq 1/4$, and*

$$D = \{y : ||y - x| - r| \leq \delta\}.$$

Let h be a complex measure whose variation is finite on $B(x, 3)$. Suppose that $K_0, K_1 : \mathbb{R}^2 \rightarrow [0, 1]$ are Borel functions supported in $B(x, 3)$, that $K_1 - K_0$ is supported in D , and that

$$\int K_i(y) \overline{y - x} dh(y) = 0 \quad (i = 0, 1).$$

Set $P_i = \int K_i dh$. For every $z \in D$,

$$P_1 - P_0 = \int \left(1 - \frac{\overline{y - x}}{z - x}\right) (K_1 - K_0)(y) dh(y). \quad (2.9)$$

For every $0 < \rho \leq 1$, this implies

$$\begin{aligned} |P_1 - P_0| &\leq 2\rho \int_{D \cap B(z, \rho)} |K_1 - K_0| d|h| \\ &\quad + 10 \int_{D \setminus B(z, \rho)} |K_1 - K_0| d|h|. \end{aligned} \quad (2.10)$$

Proof. Since $|z - x| \geq r - \delta \geq 3/4$, the denominator is nonzero. Subtract the difference of the two zero-moment identities, divided by $\overline{z - x}$, from $P_1 - P_0$. This gives (2.9). Its multiplier is

$$W_z(y) = \frac{\overline{z - y}}{\overline{z - x}}.$$

On $B(z, \rho)$ its absolute value is at most 2ρ . On D , $|y - z| \leq 2(r + \delta) \leq 9/2$ and $|z - x| \geq 3/4$, so $|W_z| \leq 6 \leq 10$. Splitting the integral into the two regions proves (2.10). $\square$

Thus concentration near z turns the factor 1 in a variation estimate into the small factor 2ρ . We will call $D \cap B(z, \rho)$ a cap of the shell. The geometric question is now concrete: when can all but a controlled amount of shell mass be placed in one cap?

For the intended use, two positive measures play different roles. One, denoted by ν , detects whether the shell carries enough error mass to require this argument. The other, denoted by S , describes positive contributions to the gradient. The conclusion must control both measures on the shell, but the available hypothesis bounds only S in the inner disk. A bound involving all of $S(\mathbb{R}^2)$ would lose that information. The tangent-disk sweep below is what allows the inner mass bound to suffice.

2.2 One exceptional set for all radii

We use Lebesgue measure for the size $|E|$ of a set of centers. The measures S and ν in the next lemma are arbitrary finite positive Borel measures; they may have atoms. No maximizing radius is selected, and no regularity of these measures is assumed.

Lemma 2.4 (Concentration in a thin shell). *Let S and ν be finite positive Borel measures on $\mathbb{R}^2$, let $U > 0$ and $T \geq 0$, and suppose that*

$$0 < \rho \leq 1, \quad 0 < \delta \leq \frac{\rho^2}{1600}.$$

Write

$$D(x, r, \delta) = \{y : ||y - x| - r| \leq \delta\}.$$

There is a Borel set $E \subset \mathbb{R}^2$ such that

$$|E| \leq C \left[\left(\frac{\nu(\mathbb{R}^2)}{U} \right)^2 \frac{\delta}{\rho} + \frac{\nu(\mathbb{R}^2)}{U} \frac{\delta}{\rho^2} \left(1 + \frac{T}{U} \right) \right], \quad (2.11)$$

and the following implication holds for every $x \notin E$ and every $r \in [1, 2]$: if

$$\nu(D(x, r, \delta)) \geq U, \quad S(B(x, r - \delta)) \leq T, \quad (2.12)$$

then there is $z \in D(x, r, \delta)$ for which

$$(S + \nu)(D(x, r, \delta) \setminus B(z, \rho)) \leq 2U. \quad (2.13)$$

The constant in (2.11) is absolute. In particular, neither the exceptional set nor its bound involves a choice of r .

Proof. If the cap conclusion fails, every shell point z sees more than U of either ν or S outside its cap. The first alternative will restrict the centers to thin strips. For the second, nested tangent disks capture that remote S -mass during a short parameter interval, starting with mass at most T . We first prove the needed geometry, then combine the two alternatives in one Borel majorant that contains no radius parameter.

Put $v = 200\delta/\rho^2$, so $v \leq 1/8$. We first record the geometry of disks tangent at a fixed point z . For a unit vector θ , the disks

$$B(z + L\theta, L), \quad L > 0,$$

increase with L . In fact, for $w = y - z$, membership is the inequality

$$|w|^2 < 2Lw \cdot \theta. \quad (2.14)$$

When $w \cdot \theta > 0$, the entry parameter is $L_*(y) = |w|^2/(2w \cdot \theta)$.

Fix x, z with $l = |x - z| \in [1/2, 3]$ and put $\theta = (x - z)/l$. If $|y - z| \geq \rho$ and $||y - x| - l| \leq 2\delta$, then, with $a = (y - z) \cdot \theta$,

$$||y - z|^2 - 2la| = ||y - x|^2 - l^2| \leq 13\delta.$$

The restriction on δ gives $a \geq \rho^2/12 > 0$, and hence

$$|L_*(y) - l| \leq 78 \frac{\delta}{\rho^2} < \frac{v}{2}. \quad (2.15)$$

Thus all such points enter strictly between $l - v$ and $l + v$. Conversely, if $|y - z| \geq \rho$ and $y \in B(z + (l - v)\theta, l - v)$, then $a > \rho^2/(2(l - v)) \geq \rho^2/6$. It follows that

$$|y - x|^2 < l^2 - 2va \leq l^2 - \frac{200}{3}\delta < (l - 2\delta)^2.$$

We have proved

$$\{y : |y - z| \geq \rho\} \cap B(z + (l - v)\theta, l - v) \subset B(x, l - 2\delta). \quad (2.16)$$

Figure 1 illustrates the nested disks and these two comparisons.

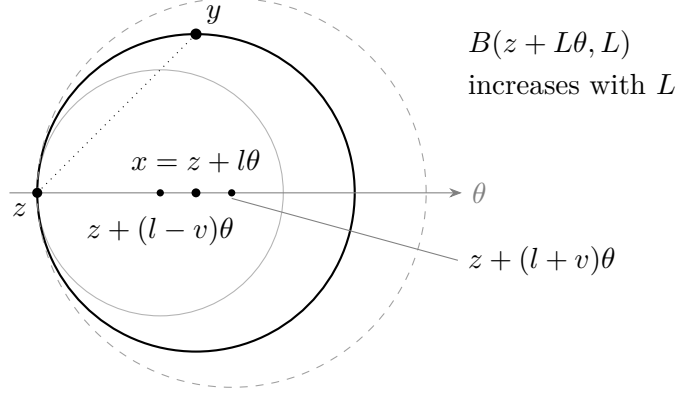


Figure 1: The tangent-disk sweep. All boundary circles pass through z . A shell point y with $|y - z| \geq \rho$ enters between the parameters $l - v$ and $l + v$, with v proportional to δ/ρ^2 . Mass outside $B(z, \rho)$ already present at $l - v$ lies in the inner disk $B(x, l - 2\delta)$. When $z \in D(x, r, \delta)$, this disk is contained in the controlled disk $B(x, r - \delta)$. The drawing is schematic.

For $z \in \mathbb{R}^2$ and θ on the unit circle define

$$H_{z,\theta}(L) = \begin{cases} S(\{y : |y - z| \geq \rho\} \cap B(z + L\theta, L)), & L > 0, \\ 0, & L \leq 0. \end{cases}$$

This is bounded and nondecreasing in L . It is jointly Borel in all its arguments, since it is the integral of the Borel indicator given by (2.14), together with the other displayed tests. For $l = |x - z| \in [1/2, 3]$ and $\theta = (x - z)/l$, let $\mathcal{C}(x, z)$ be the condition

$$H_{z,\theta}(l - v) \leq T, \quad H_{z,\theta}(l + v) - H_{z,\theta}(l - v) \geq U. \quad (2.17)$$

Define the nonnegative Borel function

$$\begin{aligned} \mathcal{M}(x) = \frac{1}{U} \int_{1/2 \leq |x-z| \leq 3} \left[\frac{1}{U} \int \mathbf{1}_{\{|y-x| - |x-z| \leq 2\delta, |y-z| \geq \rho\}} d\nu(y) \right. \\ \left. + \mathbf{1}_{\{\mathcal{C}(x,z)\}} \right] d\nu(z), \quad E = \{x : \mathcal{M}(x) \geq 1\}. \end{aligned} \quad (2.18)$$

Suppose that (2.12) holds at (x, r) and that (2.13) fails for every $z \in D = D(x, r, \delta)$. For any such z , either $\nu(D \setminus B(z, \rho)) > U$ or $S(D \setminus B(z, \rho)) > U$. Also $l = |x - z| \in [1/2, 3]$ and every $y \in D$ satisfies $||y - x| - l| \leq 2\delta$. In the first alternative, the first term in brackets in (2.18) exceeds one. In the second alternative, (2.15) gives the second inequality in (2.17). The first inequality follows from (2.16), because $l - 2\delta \leq r - \delta$. The bracket is therefore at least one for every $z \in D$. Since $\nu(D) \geq U$, we obtain $\mathcal{M}(x) \geq 1$. This proves the required implication outside the Borel set E , simultaneously for all r .

It remains to estimate $\int \mathcal{M}$. For fixed y, z with $|y - z| \geq \rho$, the centers in the first term satisfy $|x - z| \leq 3$ and

$$||x - y|^2 - |x - z|^2| \leq 13\delta.$$

The difference of squares is an affine function of x with gradient of length $2|y - z|$. These centers lie in a strip of width at most $13\delta/|y - z|$ intersected with $\overline{B}(z, 3)$, and thus have area at most $C\delta/\rho$. Tonelli's theorem bounds the first contribution to $\int \mathcal{M}$ by $C\nu(\mathbb{R}^2)^2\delta/(U^2\rho)$.

For the second contribution, fix z, θ and abbreviate $H = H_{z, \theta}$ and $G = \min(H, T + U)$. On the set of l satisfying (2.17),

$$G(l + v) - G(l - v) \geq U.$$

Because G is bounded and nondecreasing,

$$\int_{\mathbb{R}} (G(l + v) - G(l - v)) dl \leq 2v(T + U).$$

For completeness, integration over $[-M, M]$ cancels the overlapping intervals and leaves two intervals of length $2v$; letting M increase proves the inequality. Thus the permitted l have length at most $2v(1 + T/U)$. Polar coordinates about z , with $l \leq 3$, and another application of Tonelli's theorem bound the second contribution by $C\nu(\mathbb{R}^2)v(1 + T/U)/U$. Since $|E| \leq \int \mathcal{M}$, these two estimates prove (2.11). $\square$

Under its shell and inner-disk mass hypotheses, the shell lemma supplies a cap for every radius at a center outside one Borel set. The affine identity then suppresses the mass on that cap. To obtain the discounted estimates, we still have to compare kernel widths, control the positive interior mass in the directional argument, and treat radii forced at band endpoints. We do this next; the value estimate will require a separate kernel.

2.3 Cancellation of small error pieces

We now compare kernels at dyadic widths. Put

$$b = 2\alpha = \frac{1}{50}, \quad \eta = \frac{1}{10}, \quad \gamma = \frac{1}{5}, \quad \delta_j = 2^{-j}. \quad (2.19)$$

Choose a fixed $\delta_* > 0$ so small that $\delta_* \leq 1/4$ and $\delta \leq \delta^{2\gamma}/1600$ whenever $0 < \delta \leq \delta_*$.

We record estimates valid for any sequence of probability-mixture kernels K_j , centered at the same x and $r \in [1, 2]$, with the support and Lipschitz properties of Lemma 2.2 at width δ_j . The zero-moment condition is not needed for these estimates. Write

$$\nu_j = \sum_{s_d \geq \delta_j^2} |\beta_d|, \quad p_j = \int K_j d\beta, \quad D_j = D(x, r, \delta_j). \quad (2.20)$$

Then

$$\nu_j(\mathbb{R}^2) \leq A\delta_j^{-b}, \quad |p_j| \leq CA\delta_j^{-\alpha}. \quad (2.21)$$

Indeed, for $s_d < 1$, subtracting the value of K_j at the center of a supporting ball gives

$$\left| \int K_j d\beta_d \right| \leq Cm_d \min(1, s_d/\delta_j).$$

For $s_d = 1$, the same upper bound holds directly from $|K_j| \leq 1$. Now use $\min(1, a) \leq a^\alpha$ for $a > 0$ and the definition of A . The bound for ν_j follows directly from $s_d \geq \delta_j^2$.

If F is a scalar or complex test function with $\text{Lip}(F) \leq C/\delta_j$, the pieces with $s_d < \delta_j^2$ satisfy

$$\left| \sum_{s_d < \delta_j^2} \int F d\beta_d \right| \leq \frac{C}{\delta_j} \sum_{s_d < \delta_j^2} s_d m_d \leq CA\delta_j^{1-b}. \quad (2.22)$$

Only the Lipschitz bound is used here; constants cancel because each of these pieces has zero total vector mass. In particular, (2.22) applies to $K_{j+1} - K_j$ and to $\overline{y - x}(K_{j+1} - K_j)$. These tests have

the required global Lipschitz bounds, since their supports are in $B(x, 3)$. Their supports are also in D_j . Consequently

$$\begin{aligned} |p_{j+1} - p_j| &\leq C(\nu_j(D_j) + A\delta_j^{1-b}), \\ \left| \int \overline{y-x}(K_{j+1} - K_j) d\beta(y) \right| &\leq C(\nu_j(D_j) + A\delta_j^{1-b}). \end{aligned} \quad (2.23)$$

All these kernels are supported in Ω when $x \in Q$. Thus (2.2) can be used in every pairing below, even if some of its component measures have support outside Ω . For fixed x, r , as $j \rightarrow \infty$ the kernels converge off the circle to $K_\infty = \mathbf{1}_{B(x,r)}$. Absolute continuity of all the component measures gives convergence of their integrals by dominated convergence. The same holds for the moment integrals. In particular, if every finite-width kernel is balanced, then K_∞ has zero moment as well. This is a statement at each fixed (x, r) : every circle has zero mass for every absolutely continuous component. There is no exceptional set depending on the radius to remove. Absolute continuity is used here, but was not needed in the Borel shell lemma.

2.4 The directional estimate

We now prove (2.3). If $A = 0$, all the error measures vanish, and $e \cdot h(B(x, r)) \geq 0$ for every relevant disk; the conclusion follows. Assume $A > 0$.

We prove a superlevel-set estimate by separating widths at which the coarse error mass in the shell is small from those at which it is large. Small-mass widths have directly summable error increments. At the remaining widths we compare only increases of the negative projection: the later negative projection controls the interior positive mass, allowing Lemma 2.4 to be applied.

Let $\lambda \geq L_0 A$, where the fixed constant L_0 will be chosen large enough in the course of the proof. Put $q = A/\lambda$ and choose j_0 so that

$$q/2 < \delta_{j_0} \leq q. \quad (2.24)$$

We require $L_0 \geq 1/\delta_*$, so all widths under consideration are at most δ_* . At a fixed $x \in Q$ and $r \in \mathcal{I}(x)$, use Lemma 2.2 to choose K_j for every $j \geq j_0$, and write

$$P_j = \int K_j dh, \quad N_j = (-e \cdot P_j)_+, \quad U_j = \lambda \delta_j^\eta.$$

Call j active for (x, r) if $\nu_j(D_j) \geq U_j$. If it is inactive, (2.23) and $1 - b > \eta$ give

$$|p_{j+1} - p_j| + \left| \int \overline{y-x}(K_{j+1} - K_j) d\beta(y) \right| \leq C\lambda \delta_j^\eta. \quad (2.25)$$

Here and below we use $A \leq \lambda$.

Choose C_0 sufficiently large depending only on C_s and κ , and put $T_j = C_0 \lambda \delta_j^{-b}$. Apply Lemma 2.4 with

$$\nu = \nu_j, \quad \delta = \delta_j, \quad \rho = \delta_j^\gamma, \quad U = U_j, \quad T = T_j.$$

It gives Borel sets E_j , independent of r , such that

$$|E_j| \leq C \left[q^2 \delta_j^{14/25} + q \delta_j^{9/25} \right]. \quad (2.26)$$

To check the exponents, the first term of (2.11) gives $1 - \gamma - 2b - 2\eta = 14/25$. The second term is a sum of powers $1 - 2\gamma - b - \eta$ and $1 - 2\gamma - 2b - 2\eta = 9/25$; because $\delta_j \leq 1$, both are bounded by a constant times the latter. Constants in (2.26) may depend on the already fixed C_0 .

Fix a center outside $\bigcup_{j \geq j_0} E_j$, and keep any $r \in \mathcal{I}(x)$ fixed. Positivity in the direction e implies

$$e \cdot P_j \geq \kappa \int K_j dS + e \cdot p_j, \quad N_j \leq |p_j|. \quad (2.27)$$

Before the first active index, all the increments of p_j obey (2.25). Hence, if j is the first active index, or if $j = \infty$ and there are no active indices,

$$N_j \leq C \left(A \delta_{j_0}^{-\alpha} + \lambda \delta_{j_0}^\eta \right). \quad (2.28)$$

The formula at infinity follows from the dominated-convergence observation above.

The first active width has therefore been controlled. To continue, we compare consecutive active widths, allowing arbitrarily many inactive widths between them. Only an increase of the negative projection needs a new estimate; that sign will supply the positive interior-mass bound required by the shell lemma.

Suppose now that j is active and $l > j$ is the next active index; set $l = \infty$ if there is no next one. The individual bound in (2.21) at j and $j + 1$, followed by the inactive increments between $j + 1$ and l , gives

$$|p_j| + |p_l| \leq C \left(A \delta_j^{-\alpha} + \lambda \delta_j^\eta \right) \leq C \lambda \delta_j^{-b}. \quad (2.29)$$

We need to estimate a change only when $N_l > N_j$. In that case $e \cdot P_l < 0$, so (2.27) gives

$$\int K_l dS \leq \kappa^{-1} |p_l|.$$

Since $K_l = 1$ on $B(x, r - \delta_j)$, this proves $S(B(x, r - \delta_j)) \leq T_j$ after fixing C_0 large enough to dominate the constant just obtained. The shell lemma therefore supplies a point $z \in D_j$ such that

$$(S + \nu_j)(D_j \setminus B(z, \delta_j^\gamma)) \leq 2U_j. \quad (2.30)$$

This point is used only at the current center and radius; no measurable selection of z is required.

To estimate the positive mass on the cap, we need a weighted bound for both kernels being compared. The sign at l supplies one, but $e \cdot P_j > 0$ would give no corresponding bound for K_j . We therefore replace K_j , when necessary, by a kernel with zero projection and the same negative part. If $e \cdot P_j \leq 0$, set $K_* = K_j$. Otherwise, interpolate between K_j and K_l to obtain a convex combination K_* with $e \cdot P_* = 0$, where $P_* = \int K_* dh$. This is possible because $e \cdot P_l < 0$. In either case,

$$N_* := (-e \cdot P_*)_+ = N_j, \quad e \cdot P_* \leq 0.$$

Writing $p_* = \int K_* d\beta$ and using (2.29) and (2.27), we get

$$\int (K_l + K_*) dS \leq C \lambda \delta_j^{-b}. \quad (2.31)$$

All of K_j, K_l, K_* have zero moment, including when $l = \infty$ by the preceding sharp-kernel limit. Since $|z - x| \geq 1/2$, define

$$W(y) = 1 - \frac{\overline{y - x}}{\overline{z - x}}.$$

The affine identity in Lemma 2.3 gives

$$P_l - P_* = \int W(y)(K_l - K_*)(y) dh(y). \quad (2.32)$$

The difference of kernels is supported on D_j , and

$$|W| \leq C\delta_j^\gamma \quad \text{on } B(z, \delta_j^\gamma), \quad |W| \leq C \quad \text{on } D_j.$$

The contribution of $\sum_i v_i \sigma_i$ to (2.32) is therefore at most

$$C\lambda\delta_j^{\gamma-b} + CU_j, \quad (2.33)$$

by (2.31), $|K_l - K_*| \leq K_l + K_*$, and (2.30).

For the error measures, $K_l - K_*$ is a scalar multiple in $[0, 1]$ of $K_l - K_j$. Keep W fixed and telescope

$$\int W(K_l - K_j) d\beta = \sum_{i=j}^{l-1} \int W(K_{i+1} - K_i) d\beta. \quad (2.34)$$

For $l = \infty$ this equality means the limit of the finite sums. We never apply a Lipschitz cancellation bound directly to the sharp kernel K_∞ or to its interpolant: only the finite consecutive differences in this sum need such a bound. On the first step, the pieces contributing to ν_j cost at most $C\delta_j^\gamma \nu_j(\mathbb{R}^2) + CU_j$ by (2.30). The remaining pieces cost $CA\delta_j^{1-b}$ by (2.22), since $W(K_{j+1} - K_j)$ has Lipschitz constant at most C/δ_j . For $j < i < l$, use

$$\int W(K_{i+1} - K_i) d\beta = p_{i+1} - p_i - \frac{1}{z-x} \int \overline{y-x} (K_{i+1} - K_i) d\beta(y).$$

These are inactive steps, so their absolute values sum to at most $C\lambda\delta_j^\eta$ by (2.25). Combining the estimates gives

$$\left| \int W(K_l - K_j) d\beta \right| \leq C \left(A\delta_j^{\gamma-b} + A\delta_j^{1-b} + \lambda\delta_j^\eta \right).$$

Together with (2.33), and using $\gamma - b = 9/50 > \eta$, this yields

$$N_l \leq N_j + |P_l - P_*| \leq N_j + C\lambda\delta_j^\eta. \quad (2.35)$$

The same inequality is automatic when $N_l \leq N_j$.

Sum (2.35) along the active indices. If there are finitely many, include the last comparison with $l = \infty$; if there are infinitely many, pass to the limit along them. Dominated convergence and (2.28) give

$$(-e \cdot h(B(x, r)))_+ \leq C \left(A\delta_{j_0}^{-\alpha} + \lambda\delta_{j_0}^\eta \right) \leq C\lambda \left(q^{1-\alpha} + q^\eta \right). \quad (2.36)$$

This holds for every $r \in \mathcal{I}(x)$ at the fixed center outside the exceptional sets. Increase L_0 , depending only on C_s and κ , so that the last bound is less than λ whenever $q \leq L_0^{-1}$.

Let $Y(x) = \sup_{r \in \mathcal{I}(x)} (-e \cdot h(B(x, r)))_+$. Summing (2.26) and using (2.24), we have proved

$$|\{x \in Q : Y(x) > \lambda\}| \leq C \left[(A/\lambda)^{64/25} + (A/\lambda)^{34/25} \right] \quad (\lambda \geq L_0 A). \quad (2.37)$$

Both powers exceed one. The layer-cake formula, the trivial bound $|Q| = 1$ for smaller thresholds, and the substitution $\lambda = As$ give

$$\int_Q Y dx \leq L_0 A + CA \int_{L_0}^\infty (s^{-64/25} + s^{-34/25}) ds \leq CA.$$

Finally, (2.7) and $r \geq 1$ prove (2.3).

2.5 The size estimate, including band endpoints

To prove (2.4), place the measures $v_i \sigma_i$ among the error measures with assigned size one. The augmented price is $A_{\text{tot}} = A + S(\mathbb{R}^2)$, where A and S refer to the original decomposition. The augmented positive part is zero; in this proof, β and ν_j refer to the augmented error family. If $A_{\text{tot}} = 0$, the result is immediate.

Take $\lambda \geq L_0 A_{\text{tot}}$, put $q = A_{\text{tot}}/\lambda$, and choose j_0 as in (2.24). Here L_0 will depend only on C_s . For $x \in Q$ and $r \in \mathcal{J}(x)$, choose the balanced kernel K_j when $\text{dist}(r, \{1, 2\}) \geq \delta_j$. Such an r is an actual maximizer on $[1, 2]$, so the two neighboring radii needed in Lemma 2.2 are admissible. Otherwise choose the uniform probability mixture over $[r - \delta_j, r + \delta_j]$. The latter kernel need not have zero moment, but has all the same support, positivity and Lipschitz bounds. Thus (2.21)–(2.23) remain valid with A_{tot} in place of A . Write $P_j = \int K_j dh = p_j$, and again call j active when $\nu_j(D_j) \geq U_j = \lambda \delta_j^\eta$.

There is an additional Borel exceptional set for each width:

$$F_j = \bigcup_{a \in \{1, 2\}} \{x : \nu_j\{y : ||y - x| - a| \leq 2\delta_j\} \geq U_j\}. \quad (2.38)$$

For each fixed y , the areas of these two annuli of centers sum to $24\pi\delta_j$. Tonelli's theorem and Markov's inequality give

$$|F_j| \leq 24\pi\delta_j \frac{\nu_j(\mathbb{R}^2)}{U_j} \leq C(A_{\text{tot}}/\lambda)\delta_j^{22/25}. \quad (2.39)$$

If $\text{dist}(r, \{1, 2\}) < \delta_j$, the shell D_j lies in one of the annuli in (2.38). Thus, outside F_j , an active index must satisfy $\text{dist}(r, \{1, 2\}) \geq \delta_j$, and both K_j and K_{j+1} have zero moment. In particular, the forced radii $r = 1, 2$ are covered even if they are not maximizers.

Apply Lemma 2.4 with $S = 0$, $T = 0$, $\nu = \nu_j$, $\delta = \delta_j$, $\rho = \delta_j^\gamma$, and $U = U_j$. It supplies Borel exceptional sets E_j , independent of r , with bound $C[q^2\delta_j^{14/25} + q\delta_j^{12/25}]$, which is at most the right side of (2.26). Its constants have no cone parameter: the positive measure and its interior-mass threshold are both zero. At an active step outside $E_j \cup F_j$, its interior-mass hypothesis is automatic, and it gives a point $z \in D_j$ with $\nu_j(D_j \setminus B(z, \delta_j^\gamma)) \leq 2U_j$. The zero moments allow the correction

$$P_{j+1} - P_j = \int \left(1 - \frac{\overline{y-x}}{z-x}\right) (K_{j+1} - K_j)(y) d\beta(y).$$

The coarse pieces cost at most $CA_{\text{tot}}\delta_j^{\gamma-b} + CU_j$, and the fine pieces cost at most $CA_{\text{tot}}\delta_j^{1-b}$, exactly as above. Hence

$$|P_{j+1} - P_j| \leq C\lambda\delta_j^\eta.$$

At an inactive step the same inequality follows directly from (2.23), without using a moment identity. Starting with $|P_{j_0}| \leq CA_{\text{tot}}\delta_{j_0}^{-\alpha}$ and summing all increments, we obtain

$$|h(B(x, r))| \leq C\lambda(q^{1-\alpha} + q^\eta) < \lambda$$

after increasing L_0 . This holds simultaneously for all $r \in \mathcal{J}(x)$ outside $\bigcup_{j \geq j_0} (E_j \cup F_j)$. The measure bound is therefore

$$\left| \left\{ x \in Q : \sup_{r \in \mathcal{J}(x)} |h(B(x, r))| > \lambda \right\} \right| \leq C[(A_{\text{tot}}/\lambda)^{64/25} + (A_{\text{tot}}/\lambda)^{34/25} + (A_{\text{tot}}/\lambda)^{47/25}]. \quad (2.40)$$

Every exponent is larger than one. The same layer-cake computation as before gives the unnormalized bound by CA_{tot} . Since $A_{\text{tot}} = A + S(\mathbb{R}^2)$, (2.7) proves (2.4).

2.6 The value-oscillation estimate

This estimate involves all small radii and does not use a maximizing property. We express g minus a fixed mean through its gradient, prove an $L^{3/2}$ cancellation bound for the resulting kernel, and then apply the strong maximal inequality.

We finish the proof of Lemma 2.1 by establishing (2.5). Choose a nonnegative smooth probability density ψ supported in the interior of Q , with uniformly bounded derivatives after translating and rotating a fixed choice, and put

$$c = \int g(z)\psi(z) dz, \quad E_Q = \{x : \text{dist}(x, Q) < 4\}.$$

For $x \in E_Q$, integrate ∇g along the segments from x to points of $\text{supp } \psi$ and change variables to obtain

$$g(x) - c = \int L(x, y) \cdot dh(y), \quad L(x, y) = -(y - x) \int_0^1 \psi(x + (y - x)t) t^{-3} dt. \quad (2.41)$$

Indeed, this is the fundamental theorem of calculus in $g(x) - g(z) = - \int_0^1 (z - x) \cdot \nabla g(x + t(z - x)) dt$, followed by $y = x + t(z - x)$.

The displayed formula defines $L(x, y)$ for every $y \neq x$ in the plane, not only for $y \in \Omega$. This global definition will allow cancellation against a support center outside Ω . For each fixed $x \in E_Q$, the kernel is supported on segments joining x to $\text{supp } \psi$, and hence is supported in Ω . It is smooth in $y \neq x$, with the uniform bounds

$$|L(x, y)| \leq \frac{C}{|x - y|}, \quad |\nabla_y L(x, y)| \leq \frac{C}{|x - y|^2}. \quad (2.42)$$

To verify them, if the integrand or one of its derivatives is nonzero, then $|y - x| \leq Ct$, since $x \in E_Q$ and $x + (y - x)/t$ lies in a fixed bounded neighborhood of Q . The first bound follows by integrating t^{-3} from $|y - x|/C$ to one and multiplying by $|y - x|$. Differentiation in y produces either the same integral or $|y - x| \int t^{-4} dt$, which gives the second bound.

Fix $p = 3/2$, so $2/p - 1 = 1/3 > \alpha$. The first bound in (2.42) implies

$$\sup_{y \in \mathbb{R}^2} \|L(\cdot, y)\|_{L^p(E_Q)} \leq C. \quad (2.43)$$

This follows by integrating $|x - y|^{-p}$ on a bounded set; $p < 2$ makes the singularity integrable, uniformly in y .

Consider a piece β_d with $s_d < 1$, supported in $B(z, a)$, where we may take $a = C_s s_d$ after increasing C_s to at least one. On $E_Q \cap B(z, 2a)$, the first kernel bound and Minkowski's inequality give

$$\left\| \int L(\cdot, y) \cdot d\beta_d(y) \right\|_{L^p(E_Q \cap B(z, 2a))} \leq C a^{2/p-1} m_d.$$

In fact, for $y \in B(z, a)$ the relevant x lie within $3a$ of y , and the L^p norm of $|x - y|^{-1}$ there is $C a^{2/p-1}$. For $|x - z| > 2a$, use the zero total vector mass to subtract $L(x, z)$. The derivative bound in (2.42), applied on the segment from z to y , gives

$$\left| \int L(x, y) \cdot d\beta_d(y) \right| \leq C a m_d |x - z|^{-2}.$$

Its L^p norm on $\{|x - z| > 2a\}$ is at most $Ca^{2/p-1}m_d$, since $p > 1$. Combining the two regions yields

$$\left\| \int L(\cdot, y) \cdot d\beta_d(y) \right\|_{L^p(E_Q)} \leq Cs_d^{1/3}m_d \leq Cs_d^\alpha m_d. \quad (2.44)$$

The constant may depend on C_s . For a piece of assigned size one, and for the positive measures, use (2.43) directly. Minkowski's inequality and the finite decomposition therefore give

$$\|g - c\|_{L^{3/2}(E_Q)} \leq C(A + S(\mathbb{R}^2)). \quad (2.45)$$

These same estimates show that the integrals obtained by inserting (2.2) into (2.41) exist for almost every x ; thus the use of the potentially singular kernel is justified.

Let $f_Q = (g - c)\mathbf{1}_{E_Q}$, extended by zero to the plane. Every disk centered at $x \in Q$ of radius at most four is contained in E_Q . Consequently

$$\sup_{0 < t, t' \leq 4} |u_t(x) - u_{t'}(x)| \leq 2Mf_Q(x),$$

where the maximal function on the right uses absolute values. The classical strong $L^{3/2}$ maximal inequality [10, (1.2)], followed by Hölder's inequality on the unit square and (2.45), proves (2.5). This completes the proof of Lemma 2.1.

3 Dyadic vector masses and common prices

Throughout this section, let $g \in C_c^\infty(\mathbb{R}^2)$ be real valued, and write

$$h = \nabla g \, dx, \quad V = |h|(\mathbb{R}^2), \quad u_t = A_t g.$$

The averages u_t are signed. Fix an integer $m \geq 1$; all estimates below are uniform in m . Lemma 2.1 discounts a cancelling error according to the size of its support. We now decompose the gradient into errors to which that discount applies, and assign one summable price to each dyadic cell. The same price will work for every direction in a fixed finite palette. The construction starts with the actual density of each scalar coordinate of the gradient at the finest level and removes equal positive and negative masses as it moves toward coarser levels.

3.1 Positive refinement and cone coordinates

For every integer k , set $r_k = 2^{-k}$ and $V_k = r_k \mathbb{Z}^2$. These definitions are not restricted to $1, \dots, m$: the neighboring direction estimate in Section 4 will also use level $k - 12$. A vertex includes its level, so vertices at different levels are distinct even when their spatial positions agree. For $p \in V_k$, define

$$Q_p = p + [0, r_k)^2, \quad \phi_p(x) = \prod_{j=1}^2 \left(1 - \frac{|x_j - p_j|}{r_k} \right)_+.$$

For $p \in V_i$ and $q \in V_k$ with $i < k$, put $\lambda_{pq} = \phi_p(q)$, and at equal levels put $\lambda_{pq} = \mathbf{1}_{\{p=q\}}$. These nonnegative coefficients satisfy

$$\begin{aligned} \sum_{p \in V_i} \lambda_{pq} &= 1, & \phi_p &= \sum_{q \in V_k} \lambda_{pq} \phi_q \quad (i \leq k), \\ \lambda_{ps} &= \sum_{q \in V_j} \lambda_{pq} \lambda_{qs} \quad (i \leq j \leq k, \, p \in V_i, \, s \in V_k). \end{aligned} \quad (3.1)$$

To verify these identities, first consider one coordinate. The hats on a fixed grid sum to one: between two consecutive vertices, the only nonzero hats have complementary affine values. A coarse hat is affine on every interval of any finer dyadic grid, since all its breakpoints belong to that grid. It therefore equals the piecewise affine interpolant of its values at the finer vertices. Taking products proves the first two identities in (3.1); evaluating the second identity at s proves the third. The sums are locally finite. Notice that the normalization is in the coarse index p for each fixed fine vertex q . These coefficients are refinement weights, not stochastic transition rows.

If $q \in V_{k+1}$ and $\lambda_{pq} > 0$, then in each coordinate $q_j - p_j \in \{-r_k/2, 0, r_k/2\}$. Consequently

$$\text{supp } \phi_q \subseteq \text{supp } \phi_p. \quad (3.2)$$

Thus each vertex has nine positive-weight children, and the supports remain nested along every path of positive-weight edges.

Define the vector masses and their polar decompositions by

$$h_p = \int \phi_p \, dh, \quad a_p = |h_p|, \quad h_p = a_p \theta_p,$$

where θ_p is an arbitrary fixed unit vector when $a_p = 0$. Refinement gives

$$h_p = \sum_{q \in V_k} \lambda_{pq} h_q \quad (p \in V_i, i \leq k).$$

The triangle inequality and the column sums in (3.1) imply

$$\sum_{p \in V_i} a_p \leq \sum_{q \in V_k} a_q \leq V \quad (i \leq k). \quad (3.3)$$

The last inequality follows directly from $\sum_q \phi_q = 1$. At each fixed level only finitely many of these masses can be nonzero, because h has compact support.

Fix $\tau = 1/100$ and a finite $\tau/20$ -net $\mathcal{N}$ of the unit circle, measured in chord distance. For $n \in \mathcal{N}$, let $n^\perp$ be its counterclockwise quarter turn and put

$$e_+(n) = \frac{\sqrt{3}}{2}n + \frac{1}{2}n^\perp, \quad e_-(n) = \frac{\sqrt{3}}{2}n - \frac{1}{2}n^\perp.$$

These are unit vectors and $e_+(n) \cdot e_-(n) = 1/2$. Call (q, n) *good* when θ_q belongs to the closed sector between $e_-(n)$ and $e_+(n)$, and *bad* otherwise. The two scalar coefficient functionals

$$c_{n,+}(z) = \frac{n \cdot z}{\sqrt{3}} + n^\perp \cdot z, \quad c_{n,-}(z) = \frac{n \cdot z}{\sqrt{3}} - n^\perp \cdot z \quad (3.4)$$

satisfy

$$z = e_+(n)c_{n,+}(z) + e_-(n)c_{n,-}(z), \quad |c_{n,+}(z)| + |c_{n,-}(z)| = 2 \max \left\{ \frac{|n \cdot z|}{\sqrt{3}}, |n^\perp \cdot z| \right\} \leq 2|z|.$$

In particular, for a good pair (q, n) , both coefficients of h_q are nonnegative, including when $h_q = 0$. This sign information will supply the positive cone measures in Lemma 2.1.

For the finite-band assembly, we also need to bound a vector's norm by its signed projection along n and the unfavorable projections controlled by the local estimate. The following inequality gives that bound, with an absolute constant:

$$|z| \leq C \left(n \cdot z + \sum_{\varepsilon \in \{+, -\}} (-e_\varepsilon(n) \cdot z)_+ \right) \quad (z \in \mathbb{R}^2). \quad (3.5)$$

Here the expression in parentheses is nonnegative. Indeed, writing $a = n \cdot z$, its value is positive for every nonzero z : if $a > 0$ this is immediate; if $a = 0$, one of the two negative parts equals $|z|/2$; and if $a < 0$, their sum is at least $-(e_+(n) + e_-(n)) \cdot z = \sqrt{3}|a|$, so the expression is at least $(\sqrt{3} - 1)|a| > 0$. Its minimum on the unit circle is therefore positive by continuity. Rotation invariance makes this minimum independent of n , and homogeneity proves (3.5).

3.2 Cancellation from the finest level upward

Let $\mathcal{C}$ be the indexed finite family consisting of $c_{n,+}, c_{n,-}$ for every $n \in \mathcal{N}$; repeated functionals, if any, retain their indices. For $c \in \mathcal{C}$ write

$$\nu^c = c(h) = c(\nabla g(x)) \, dx, \quad c_s = c(h_s) = \int \phi_s \, d\nu^c.$$

For $s \in V_l$, $1 \leq l \leq m$, define

$$d_s^c = \begin{cases} \frac{1}{2} \left(\sum_{t \in V_{l+1}} \lambda_{st} |c_t| - |c_s| \right), & l < m, \\ \frac{1}{2} \left(\int \phi_s(x) |c(\nabla g(x))| \, dx - |c_s| \right), & l = m. \end{cases} \quad (3.6)$$

These numbers are nonnegative by refinement and the triangle inequality. They measure the equal positive and negative masses that can be removed at s . At level m the definition uses the positive and negative parts of the density itself. In particular, $c_s = 0$ need not imply $d_s^c = 0$.

We will realize this numerical cancellation by measures supported in the corresponding tent. The elementary operation is proportional removal; it does not require the two measures being paired to be mutually singular.

Lemma 3.1 (Removing equal masses). *Let U, V be finite nonnegative measures, of masses P, N . There are nonnegative submeasures $\mu_+ \leq U$, $\mu_- \leq V$, both of mass $d = \min(P, N)$, such that*

$$U - V = \rho + \mu_+ - \mu_-, \quad \rho(\mathbb{R}^2) = P - N, \quad |\rho|(\mathbb{R}^2) = |P - N|.$$

The residual ρ has constant sign. If U, V are absolutely continuous and supported in a common closed set, so are all the constructed measures.

Proof. When $P > 0$ set $\mu_+ = (d/P)U$; when $P = 0$ set $\mu_+ = 0$. Define μ_- in the same way from N, V , and set $\rho = (U - \mu_+) - (V - \mu_-)$. If $P \geq N$ and $P > 0$, then $\mu_- = V$ and $\rho = (1 - N/P)U \geq 0$. The case $N \geq P$ is symmetric. If $P = N = 0$, every measure is zero. These formulas prove the mass, sign and support assertions, including all zero cases. $\square$

Lemma 3.2 (Exact scalar pairing). *For every $c \in \mathcal{C}$ there are measures $\rho_s^c, \mu_{s,+}^c, \mu_{s,-}^c$ at all vertices $s \in V_l$, $1 \leq l \leq m$, supported in $\text{supp } \phi_s$ and absolutely continuous, with ρ_s^c of constant sign and $\mu_{s,\pm}^c$ nonnegative, such that*

$$\rho_s^c(\mathbb{R}^2) = c_s, \quad |\rho_s^c|(\mathbb{R}^2) = |c_s|, \quad \mu_{s,+}^c(\mathbb{R}^2) = \mu_{s,-}^c(\mathbb{R}^2) = d_s^c. \quad (3.7)$$

Put $\beta_s^c = \mu_{s,+}^c - \mu_{s,-}^c$. For every $q \in V_k$, $1 \leq k \leq m$, the following is an identity of measures on the whole plane:

$$\phi_q \nu^c = \rho_q^c + \sum_{l=k}^m \sum_{s \in V_l} \lambda_{qs} \beta_s^c. \quad (3.8)$$

Only finitely many terms in this identity are nonzero, and

$$\beta_s^c(\mathbb{R}^2) = 0, \quad |\beta_s^c|(\mathbb{R}^2) \leq 2d_s^c, \quad \text{supp } \beta_s^c \subseteq \text{supp } \phi_s \subseteq \overline{B}(s, \sqrt{2}r_l). \quad (3.9)$$

Proof. Fix c and first take $s \in V_m$. Apply Lemma 3.1 to the actual terminal measures

$$U_s = \phi_s(\nu^c)_+, \quad V_s = \phi_s(\nu^c)_-.$$

Their masses have difference c_s and sum $\int \phi_s d|\nu^c|$. Hence their smaller mass is exactly d_s^c , and removal gives (3.7) and

$$\phi_s \nu^c = \rho_s^c + \beta_s^c. \quad (3.10)$$

If the positive and negative terminal masses agree, the residual is zero while both paired measures keep that common mass.

Suppose that all residuals at level $l+1$ have been constructed. At $s \in V_l$ combine their positive and negative parts separately:

$$U_s = \sum_{t \in V_{l+1}} \lambda_{st}(\rho_t^c)_+, \quad V_s = \sum_{t \in V_{l+1}} \lambda_{st}(\rho_t^c)_-.$$

The residuals have constant sign, so the masses P_s, N_s of these two measures obey

$$P_s - N_s = \sum_t \lambda_{st} c_t = c_s, \quad P_s + N_s = \sum_t \lambda_{st} |c_t|.$$

Thus $\min(P_s, N_s) = d_s^c$. Applying Lemma 3.1 again constructs the measures in (3.7) and proves

$$\sum_{t \in V_{l+1}} \lambda_{st} \rho_t^c = \rho_s^c + \beta_s^c. \quad (3.11)$$

Absolute continuity is preserved by finite sums and scalar multiplication. Support nesting in (3.2) keeps all these measures in $\text{supp } \phi_s$. The construction proceeds through finitely many levels. At each level only finitely many measures can be nonzero, since the terminal density has compact support and each vertex has finitely many parents.

It remains to verify that no density has been lost during removal. At level m , (3.8) is (3.10). If the identity holds for all children of $q \in V_k$, refinement yields

$$\begin{aligned} \phi_q \nu^c &= \sum_{t \in V_{k+1}} \lambda_{qt} \phi_t \nu^c \\ &= \sum_t \lambda_{qt} \rho_t^c + \sum_{l=k+1}^m \sum_{s \in V_l} \left(\sum_t \lambda_{qt} \lambda_{ts} \right) \beta_s^c \\ &= \rho_q^c + \beta_q^c + \sum_{l=k+1}^m \sum_{s \in V_l} \lambda_{qs} \beta_s^c. \end{aligned}$$

The last line uses (3.11) and coefficient composition in (3.1). It is the required identity at q . All rearrangements are finite for a fixed root because branching and depth are finite. Finally, equal masses in (3.7) give zero mass and variation at most $2d_s^c$ for each difference; the tent support gives the last assertion in (3.9). $\square$

The measures are constructed once at each vertex. A descendant's pair may appear in several coarse identities, with its specified coefficient; these identities do not assert a disjoint allocation.

The equivalent path form. The price estimate below uses the vertex identity. The following equivalent form expresses shared-descendant coefficients as sums of path weights, using the same cancelling pieces. Section 4 will construct a separate positive measure on vector paths. Fix $q \in V_k$ with $1 \leq k \leq m$, and let ξ range over all positive-weight paths $\xi = (q_k, \dots, q_l)$ with $q_k = q$ and $k \leq l \leq m$, including the trivial path (q) . Set

$$w_\xi = \prod_{j=k}^{l-1} \lambda_{q_j q_{j+1}}, \quad \mu_{\xi, \pm}^c = w_\xi \mu_{q_l, \pm}^c, \quad w_{(q)} = 1.$$

Composition gives $\sum_{\xi: q_l=s} w_\xi = \lambda_{qs}$, and therefore

$$\phi_q^c(h) = \rho_q^c + \sum_{\xi} (\mu_{\xi, +}^c - \mu_{\xi, -}^c). \quad (3.12)$$

Each path pair has mass $w_\xi d_s^c$ in each sign and support in $\text{supp } \phi_s$, where $s = q_l$. This recovers the path identity without a separate allocation along every path.

The cancellation amounts have an exact total budget. Summing (3.6) over a level and using the column normalization in (3.1) cancels intermediate absolute masses. The terminal partition of unity gives

$$\sum_{l=1}^m \sum_{s \in V_l} d_s^c = \frac{1}{2} \left(\int |c(\nabla g)| \, dx - \sum_{s \in V_1} |c_s| \right) \leq \frac{1}{2} \|c\| V. \quad (3.13)$$

Here $\|c\|$ denotes the Euclidean operator norm, equal to $2/\sqrt{3}$ for every $c \in \mathcal{C}$ by (3.4). When $m = 1$, the same formula is just the sum of the terminal defects. Zero masses introduce no exception to either the reconstruction or this telescoping identity.

3.3 One price at every observation scale

For $Q = Q_p$ at level k , put

$$\mathcal{R}(Q) = \{q \in V_k : |q - p|_\infty \leq 20r_k\}, \quad |z|_\infty = \max\{|z_1|, |z_2|\}.$$

A pair created at level $l \geq k$ has support of radius at most $\sqrt{2}r_l$. At observation scale r_k , the local estimate therefore charges it the relative factor $(r_l/r_k)^\alpha = 2^{-\alpha(l-k)}$. We sum these charges over all the scalar coordinates so that the resulting price is independent of the label chosen later.

The scaled radius sets from Lemma 2.1 are

$$\begin{aligned} \mathcal{I}_k(x) &= \{r \in [r_k, 2r_k] : u_r(x) \geq u_t(x) \text{ for every } t > 0 \text{ with } |t - r| \leq r_k/4\}, \\ \mathcal{J}_k(x) &= \{r \in [r_k, 2r_k] : u_r(x) = \max_{r_k \leq t \leq 2r_k} u_t(x)\} \cup \{r_k, 2r_k\}. \end{aligned}$$

Also define

$$v_k(x) = \sup_{0 < t, t' \leq 4r_k} |u_t(x) - u_{t'}(x)|.$$

An empty supremum of nonnegative quantities is zero, as before.

Proposition 3.3 (Common discounted prices). *For $p \in V_k$, $1 \leq k \leq m$, define*

$$D_{Q_p} = \sum_{c \in \mathcal{C}} \sum_{q \in \mathcal{R}(Q_p)} \sum_{l=k}^m 2^{-\alpha(l-k)} \sum_{s \in V_l} \lambda_{qs} d_s^c, \quad \alpha = \frac{1}{100}. \quad (3.14)$$

These nonnegative prices satisfy

$$\sum_{k=1}^m \sum_{p \in V_k} D_{Q_p} \leq \frac{41^2}{2(1-2^{-\alpha})} \sum_{c \in \mathcal{C}} \|c(\nabla g)\|_{L^1} \leq CV. \quad (3.15)$$

For every such cell Q and every $n \in \mathcal{N}$,

$$\begin{aligned} \int_Q \sup_{t \in \mathcal{I}_k(x)} \sum_{\varepsilon \in \{+, -\}} (-e_\varepsilon(n) \cdot \nabla u_t(x))_+ \, dx &\leq C \left(D_Q + \sum_{\substack{q \in \mathcal{R}(Q) \\ (q,n) \text{ bad}}} a_q \right), \\ \int_Q \left[\sup_{t \in \mathcal{J}_k(x)} |\nabla u_t(x)| + r_k^{-1} v_k(x) \right] dx &\leq C \left(D_Q + \sum_{q \in \mathcal{R}(Q)} a_q \right). \end{aligned} \quad (3.16)$$

All constants are absolute: the direction palette, support radius factor and cone aperture are fixed, and there is no dependence on m .

Proof. We first sum the prices. At a fixed level each vertex q belongs to exactly 41^2 neighborhoods $\mathcal{R}(Q_p)$, since each of the two integer coordinate differences ranges from -20 to 20 . Nonnegativity permits reordering, and the column sums give

$$\begin{aligned} \sum_{k=1}^m \sum_{p \in V_k} D_{Q_p} &= 41^2 \sum_{c \in \mathcal{C}} \sum_{l=1}^m \sum_{s \in V_l} d_s^c \sum_{k=1}^l 2^{-\alpha(l-k)} \sum_{q \in V_k} \lambda_{qs} \\ &\leq \frac{41^2}{1-2^{-\alpha}} \sum_{c \in \mathcal{C}} \sum_{l=1}^m \sum_{s \in V_l} d_s^c. \end{aligned}$$

Applying the exact telescoping identity (3.13) proves (3.15). In particular, one may take $C = 41^2 |\mathcal{C}| / [\sqrt{3}(1-2^{-\alpha})]$ in its final inequality.

We next construct the decomposition needed by the local estimates. Fix $Q = Q_p$ at level k , a label n , and write $\Omega_Q = \{y : \text{dist}(y, Q) < 10r_k\}$. If a level- k tent ϕ_q is positive at $y \in \Omega_Q$, then $|q - y|_\infty < r_k$ and $|y - p|_\infty < 11r_k$, so $q \in \mathcal{R}(Q)$. Hence

$$h = \sum_{q \in \mathcal{R}(Q)} \phi_q h \quad \text{on } \Omega_Q. \quad (3.17)$$

Apply (3.8) with the two coordinates $c_{n,\varepsilon}$ and multiply by $e_\varepsilon(n)$. This gives the measure identity

$$h = \sum_{q \in \mathcal{R}(Q)} \sum_{\varepsilon \in \{+, -\}} e_\varepsilon(n) \rho_q^{c_{n,\varepsilon}} + \sum_{q \in \mathcal{R}(Q)} \sum_{\varepsilon \in \{+, -\}} \sum_{l=k}^m \sum_{s \in V_l} e_\varepsilon(n) \lambda_{qs} \beta_s^{c_{n,\varepsilon}} \quad \text{on } \Omega_Q. \quad (3.18)$$

The components here are measures on the whole plane. We retain them on their full supports; only their sum is being compared with h on Ω_Q .

For a good root (q, n) , both $c_{n,\varepsilon}(h_q)$ are nonnegative. The constant-sign and mass assertions in (3.7) therefore imply $\rho_q^{c_{n,\varepsilon}} \geq 0$, including the zero case. These residuals are the positive measures σ_i of Lemma 2.1, in the unit directions $e_+(n), e_-(n)$. Each of these directions has dot product at least $1/2$ with either test direction $e_+(n)$ or $e_-(n)$. For a bad root, put the two signed residual terms among the unrestricted errors, with assigned relative size one. Their total variation is at most

$$\sum_{\varepsilon \in \{+, -\}} |\rho_q^{c_{n,\varepsilon}}|(\mathbb{R}^2) = \sum_{\varepsilon \in \{+, -\}} |c_{n,\varepsilon}(h_q)| \leq 2a_q.$$

Thus no good-root mass enters the error price for the directional estimate.

For each term in the second sum of (3.18), its total vector mass is zero, its variation is at most $2\lambda_{q_s} d_s^{c_{n,\varepsilon}}$, and its support lies in $\overline{B}(s, \sqrt{2}r_l)$. Assign it the relative size $s_d = r_l/r_k = 2^{k-l}$. When $l > k$ this gives exactly the cancellation and support conditions required for a small error. When $l = k$ it gives an allowed size-one error. The physical sum of discounted variations is consequently bounded by

$$\sum_d s_d^\alpha |\beta_d|(\mathbb{R}^2) \leq 2D_Q + 2 \sum_{\substack{q \in \mathcal{R}(Q) \\ (q,n) \text{ bad}}} a_q. \quad (3.19)$$

Here β_d denotes the vector error pieces just specified. The sum is finite and every component is absolutely continuous. The terminal pairs are included, even at roots with zero vector mass; there is no terminal remainder.

To apply Lemma 2.1, let $r = r_k$, $T(X) = p + rX$ and $\tilde{g} = g \circ T$. Its gradient measure and disk averages satisfy

$$\tilde{h} = r^{-1}(T^{-1})_\# h, \quad A_t \tilde{g}(X) = u_{rt}(T(X)), \quad \nabla_X A_t \tilde{g}(X) = r \nabla u_{rt}(T(X)). \quad (3.20)$$

Indeed $\nabla_X \tilde{g} = r \nabla g \circ T$ and $dX = r^{-2} dx$. Transform every component measure by the same $r^{-1}(T^{-1})_\#$ rule. Its mass and variation are multiplied by r^{-1} ; zero total vector mass and positivity are preserved. A pair at level l is now supported in a ball of radius $\sqrt{2}r_l/r_k$. Taking $C_s = 2$ places this closed support inside an open ball of radius $C_s r_l/r_k$. The transformed identity holds on the 10-neighborhood of $[0, 1)^2$.

The radius sets for $\tilde{g}$ are the images of $\mathcal{I}_k, \mathcal{J}_k$ under division by r , including the comparison with all positive radii in $\mathcal{I}_k$ and both forced endpoints in $\mathcal{J}_k$. Furthermore, the integral over the unit cell of any of the gradient suprema in Lemma 2.1 is r^{-1} times the corresponding physical integral over Q . Apply (2.3) with cone margin $\kappa = 1/2$ to each of the two test directions, and multiply by r . The supremum of the sum of the two nonnegative terms is at most the sum of their suprema. Together with (3.19), this proves the first estimate in (3.16).

For the second estimate in (3.16), fix any one label and use the same decomposition, now placing *all* root residuals among the size-one errors. The positive part S is zero, and the physical discounted error mass is at most $2D_Q + 2 \sum_{q \in \mathcal{R}(Q)} a_q$. The size estimate (2.4), after the same transformation, bounds the physical gradient integral by an absolute multiple of this quantity. For the value estimate the change of variables instead gives

$$\int_{[0,1]^2} \sup_{0 < t, t' \leq 4} |A_t \tilde{g}(X) - A_{t'} \tilde{g}(X)| dX = r^{-2} \int_Q v_k(x) dx. \quad (3.21)$$

Its right side in (2.5) is bounded by $Cr^{-1}(D_Q + \sum_q a_q)$. Multiplying by r therefore bounds $r_k^{-1} \int_Q v_k$, exactly as asserted. Adding this to the size bound completes the proof. $\square$

The prices now pay for every scalar cancellation and leave only nearby vector masses a_q in the local bounds. Controlling those remaining masses requires a choice of directions across scales and positions. The next section constructs a positive measure on complete refinement paths, recording a birth level, so that the mass of paths through a vertex q whose birth has already occurred is exactly a_q . The scalar measures above supply neither this path measure nor the spatial estimates for the direction law; their role is the reconstruction and discounted budget proved in this section.

4 Directions that change slowly in scale and position

The common prices in Proposition 3.3 pay for scalar cancellation. Its remaining terms are nearby vector masses a_q . We now choose directions so that these masses pay only a summable amount for changes of direction. A maximal consecutive interval of levels carrying the same direction will be called a *run*. Ending a run produces endpoint terms in the local estimates, so the first cost is the probability of a change between successive levels.

There is also a spatial cost. We will average signed directional estimates with weights depending on x and then integrate by parts. For any finite Lipschitz probability row $(\omega_j(x))_j$, normalization gives

$$\sum_j \nabla \omega_j = 0, \quad \sum_j U_j \nabla \omega_j = \sum_j (U_j - U_*) \nabla \omega_j$$

almost everywhere, for arbitrary scalar values U_j and a common value U_* . Thus the cost of differentiating a row is controlled by the differences $U_j - U_*$, rather than the sizes of the values themselves. In Section 5, the differences will be bounded by sums of the small-radius oscillations v_l defined in Section 3. Their integral over a level- l cell costs r_l times its price and root masses, by (3.16). A row built from level- k tents has derivatives of size r_k^{-1} . The required joint bound therefore carries the ratio $r_l/r_k = 2^{k-l}$, for $k \leq l$. This explains both the row-derivative cost and its weight in the packing estimate below.

Retain $g \in C_c^\infty(\mathbb{R}^2)$, $h = \nabla g \, dx$, $V = |h|(\mathbb{R}^2)$, and the levels $1, \dots, m$, with $m \geq 1$. We use the hats, vector masses $h_q = a_q \theta_q$, and refinement coefficients from Section 3, defined at every integer level. The coarser levels will be needed to compare nearby directions. The palette $\mathcal{N}$ is the fixed $\tau/20$ -net of the unit circle, where $\tau = 1/100$. Recall that (q, n) is good when θ_q makes an angle at most $\pi/6$ with n . All vertex and palette choices below use fixed tie rules, independently of the observation point x . Directions at zero vector masses retain the prescribed unit-vector convention.

4.1 Directions in neighboring cells

At each vertex we first look for a nearby direction carrying maximal mass. This choice prevents a small neighboring mass from imposing an unpaid change of direction. For $z \in V_k$, choose a vertex $w(z) \in V_k$ maximizing a_w among $|w - z|_\infty \leq 300r_k$, and set

$$\Theta_z = \theta_{w(z)}.$$

Choose $n_z \in \mathcal{N}$ with $|n_z - \Theta_z| \leq \tau/20$. The level is part of the vertex label in this notation. Define

$$E_k(q) = \max_{\substack{z \in V_k \\ |z - q|_\infty \leq 200r_k}} |\Theta_z - \theta_q|^2, \quad q \in V_k.$$

In particular, $0 \leq E_k(q) \leq 4$. The quantity $E_k(q)$ measures how far the nearby chosen directions can depart from its own direction. Put $A_j = \sum_{p \in V_j} a_p$ for every integer j . By (3.3), these total vector masses satisfy $0 \leq A_j \leq A_{j+1} \leq V$. Their increase pays the weighted sum of the disagreements.

Lemma 4.1 (Neighboring direction budget). *There is an absolute constant C such that*

$$\sum_{k=1}^m \sum_{q \in V_k} a_q E_k(q) \leq CV. \tag{4.1}$$

Proof. Fix k and q with $a_q > 0$, choose a maximizing z in the definition of $E_k(q)$, and put $w = w(z)$. Since q is among the candidates for $w(z)$, we have

$$a_w \geq a_q, \quad |w - q|_\infty \leq 500r_k.$$

Take $L = 12$, and choose a nearest vertex $p \in V_{k-L}$ to q in each coordinate. Then

$$|q - p|_\infty \leq \frac{1}{2}r_{k-L}, \quad |w - p|_\infty \leq \left(\frac{1}{2} + 500 \cdot 2^{-12}\right)r_{k-L} < \frac{5}{8}r_{k-L}.$$

Each coordinate factor of $\phi_p(q)$ is at least $1/2$, and each coordinate factor of $\phi_p(w)$ is greater than $3/8$. Hence $\lambda_{pq} \geq 1/4$ and $\lambda_{pw} \geq 9/64$. For this coarse vertex write

$$H_{p,k} = \sum_{s \in V_k} \lambda_{ps} a_s - a_p.$$

The vector refinement identity implies, also when $a_p = 0$,

$$H_{p,k} = \sum_{s \in V_k} \lambda_{ps} a_s (1 - \theta_p \cdot \theta_s) = \frac{1}{2} \sum_{s \in V_k} \lambda_{ps} a_s |\theta_s - \theta_p|^2. \quad (4.2)$$

Here $\theta_p \cdot h_p = a_p$ is true even at zero, for the prescribed unit direction at zero. The triangle inequality for squared distances, the two lower bounds on the interpolation coefficients, and $a_w \geq a_q$ therefore give

$$a_q E_k(q) \leq 2a_q (|\theta_q - \theta_p|^2 + |\theta_w - \theta_p|^2) \leq C H_{p,k}.$$

For fixed k , each coarse vertex is chosen by at most a fixed number of fine vertices, depending only on L . Vertices with $a_q = 0$ contribute nothing. The column sums of the refinement coefficients now give

$$\sum_{q \in V_k} a_q E_k(q) \leq C \sum_{p \in V_{k-L}} H_{p,k} = C(A_k - A_{k-L}).$$

Finally,

$$\sum_{k=1}^m (A_k - A_{k-L}) = \sum_{k=1}^m \sum_{j=k-L+1}^k (A_j - A_{j-1}) \leq LV,$$

because every nonnegative increment appears at most L times. This proves (4.1). $\square$

4.2 The finite-state label law

The neighboring budget does not prescribe a direction at every point. We interpolate the fixed vertex choices by the tents, retaining the previous direction unless it is appreciably far from the sampled one. The interval on which the switching probability vanishes will keep small direction fluctuations from creating repeated run endpoints.

For each $x \in \mathbb{R}^2$ we define a probability law $\mathbb{P}_x$ on $\mathcal{N}^m$. At the first step, sample $z \in V_1$ with probabilities $\phi_z(x)$ and set $n_1 = n_z$. At a step $k \geq 2$, condition on the old label $n = n_{k-1}$, sample $z \in V_k$ with probabilities $\phi_z(x)$, and switch to n_z with probability

$$p_k(n, z) = P(|n - \Theta_z|), \quad P(d) = \min\{1, \tau^{-2}(d - \tau)_+^2\}. \quad (4.3)$$

Otherwise retain n . These samples and switch trials are made independently of the past conditional on the old label. If $p_k(n, z) > 0$, then $n_z \neq n$: the opposite equality would imply $|n - \Theta_z| \leq \tau/20 < \tau$.

For a label n , let δ_n denote the probability vector concentrated at n . The initial row and subsequent transition rows are, respectively,

$$\pi_1(x) = \sum_{z \in V_1} \phi_z(x) \delta_{n_z}, \quad (4.4)$$

$$\pi_k(x; n, \cdot) = \delta_n + \sum_{z \in V_k} \phi_z(x) p_k(n, z) (\delta_{n_z} - \delta_n), \quad k \geq 2. \quad (4.5)$$

Every entry is nonnegative and every row sums to one, as is also clear from the sampling rule. The rows are continuous, piecewise bilinear interpolations of bounded vectors, with Lipschitz constant at most C/r_k . These formulas, together with the initial row, specify the probability of each label sequence as a product of entries; this is the law $\mathbb{P}_x$. We denote its expectation by $\mathbb{E}_x$ and the marginal law of n_k by $\zeta_k(x)$. When convenient, the initial row is viewed as a transition from a single auxiliary state $*$, with $\zeta_0(x) = \delta_*$ for every x .

The row switch probability for a prescribed old label is

$$\rho_k(x | n) = \sum_{z \in V_k} \phi_z(x) p_k(n, z), \quad k \geq 2.$$

Since a trial counted by this expression changes the label, define

$$\begin{aligned} R_k(x) &= \sum_{n \in \mathcal{N}} \zeta_{k-1}(x; n) \rho_k(x | n) = \mathbb{P}_x(n_k \neq n_{k-1}), \quad 2 \leq k \leq m, \\ R_1(x) &= R_{m+1}(x) = 1. \end{aligned} \quad (4.6)$$

The two sentinel values count the beginning and end of a history; they apply also when $m = 1$. The functions R_k are continuous, since the rows are Lipschitz and each marginal is obtained by finitely many matrix products.

Let G be the union of the grid lines at levels $1, \dots, m$. It is a fixed Lebesgue-null set. Off G all the tent rows are differentiable. Define the scaled expected row-derivative costs there by

$$\begin{aligned} T_k(x) &= r_k \sum_{n \in \mathcal{N}} \zeta_{k-1}(x; n) \sum_{n' \in \mathcal{N}} |\nabla_x \pi_k(x; n, n')|, \quad 2 \leq k \leq m, \\ T_1(x) &= r_1 \sum_{n' \in \mathcal{N}} |\nabla_x \pi_1(x; n')|. \end{aligned} \quad (4.7)$$

Set $T_k = 0$ on G . These are measurable versions of the costs, and all later bounds on their suprema are essential-supremum bounds. The derivative in this definition acts on one transition row; it is not the derivative of the marginal ζ_k . Off G , at most four level- k tents have a nonzero gradient, and $|\nabla \phi_z| \leq \sqrt{2}/r_k$. It follows from the row formulas that

$$0 \leq R_k \leq 1, \quad 0 \leq T_k \leq C. \quad (4.8)$$

The norm inside the inner sum in (4.7) is the Euclidean norm of the spatial gradient, whereas norms of probability vectors below are their ℓ^1 norms.

Lemma 4.2 (Good labels near a controlled vertex). *Set $c_* = (\tau/10)^2$. Suppose $q \in V_k$, $E_k(q) \leq c_*$, and $|x - q|_\infty \leq 50r_k$. Then (q, n_k) is good with $\mathbb{P}_x$ -probability one.*

Proof. Every vertex z with $\phi_z(x) > 0$ satisfies $|z - q|_\infty < 51r_k < 200r_k$, and hence $|\Theta_z - \theta_q| \leq \tau/10$. An initial or switched label therefore has distance at most $\tau/10 + \tau/20$ from θ_q . If a label n is retained after sampling z with positive probability, then $P(|n - \Theta_z|) < 1$, so $|n - \Theta_z| < 2\tau$. Its distance from θ_q is consequently less than $21\tau/10$. Both bounds are less than $2\sin(\pi/12)$, since $\tau = 1/100$. The angle between the resulting label and θ_q is thus less than $\pi/6$, as required by the definition of a good pair. $\square$

This lemma pays the bad-root terms in the first line of (3.16). If $q \in \mathcal{R}(Q_p)$ at level k , then $|x - q|_\infty \leq 21r_k$ throughout Q_p . Whenever (q, n) is bad and $E_k(q) \leq c_*$, the marginal probability of $n_k = n$ therefore vanishes on that whole cell. In the remaining case, $a_q \leq c_*^{-1} a_q E_k(q)$, whose sum is bounded by (4.1), with the fixed cell and palette multiplicities. It remains to pay root masses at the endpoints of runs and in the terms where spatial weights are differentiated.

For $q \in V_l$ define the closed square

$$D(q) = \{x \in \mathbb{R}^2 : |x - q|_\infty \leq 25r_l\}.$$

It contains every level- l pricing cell whose root neighborhood contains q : indeed $q \in \mathcal{R}(Q_p)$ and $x \in Q_p$ give $|x - q|_\infty \leq 21r_l$. We can now state the estimate that pays both types of cost on those whole cells.

Proposition 4.3 (Packing of switch and derivative costs). *For every real $g \in C_c^\infty(\mathbb{R}^2)$ and every $m \geq 1$, the quantities constructed above satisfy*

$$\sum_{l=1}^m \sum_{q \in V_l} a_q \left[\sup_{x \in D(q)} (R_l(x) + R_{l+1}(x)) + \sum_{k=1}^l 2^{k-l} \operatorname{ess\,sup}_{x \in D(q)} T_k(x) \right] \leq CV. \quad (4.9)$$

The constant is absolute and independent of m .

We prove this proposition in three steps. First a positive measure on refinement paths will realize each a_q exactly. Along each path, vector additivity will bound switches at its terminal position. Finally comparison of the laws at nearby points will turn that fixed-position estimate into the suprema over $D(q)$.

4.3 Particles and intrinsic direction changes

The scalar decomposition in Section 3 is an identity of signed measures with cancelling pairs. Here we need a different object: a positive measure on complete refinement paths, with exact occupancy a_q after birth. It must also retain the path before birth, because the label law has been evolving since level one. This path measure will not depend on x . We will combine it with $\mathbb{P}_x$ by evaluating that law at a path's terminal vertex. A path node at level k is $\xi = (q_1, \dots, q_k)$, where $q_j \in V_j$ and each consecutive refinement coefficient is positive. Its path weight and mass are

$$b_\xi = \prod_{j=1}^{k-1} \lambda_{q_j q_{j+1}}, \quad M_\xi = b_\xi a_{q_k}.$$

The empty product at a root is one. The associated vector is $b_\xi h_{q_k}$. Vectors add from the children to their parent, and therefore

$$M_\xi \leq \sum_{\eta \text{ child of } \xi} M_\eta. \quad (4.10)$$

Only a finite forest is needed. Choose all level-one roots whose closed tent supports meet $\operatorname{supp} h$, and retain all their positive-weight paths through level m . There are finitely many roots and each node has finitely many children. If $a_q > 0$, the support of ϕ_q meets $\operatorname{supp} h$. Every ancestor tent then meets $\operatorname{supp} h$ by support nesting, so every positive-weight path to q starts among the chosen roots. In particular no contribution to a nonzero mass is lost by this finite restriction. If $h = 0$, the forest may be empty. Consequently, at each level k ,

$$\sum_{\xi: q_k = q} b_\xi = 1 \quad \text{if } a_q > 0, \quad \sum_{|\xi|=k} M_\xi = A_k. \quad (4.11)$$

Lemma 4.4 (A finite particle measure). *There is a finite positive measure μ on pairs consisting of a terminal path $(q_1, \dots, q_m)$ and a birth level $B \in \{1, \dots, m\}$ such that, for every node ξ at level k ,*

$$\mu\{B \leq k, (q_1, \dots, q_k) = \xi\} = M_\xi. \quad (4.12)$$

In particular,

$$\mu\{B \leq k, q_k = q\} = a_q, \quad \mu(\text{all particles}) = A_m \leq V. \quad (4.13)$$

Writing $y = q_m$ for the terminal position of a particle, its complete path obeys

$$|y - q_s|_\infty < r_s \quad (1 \leq s \leq m). \quad (4.14)$$

The path is defined also at levels preceding its birth.

Proof. At each root create mass M_ξ with birth level one. Suppose the population already present at a level- k node has mass M_ξ , and put $S_\xi = \sum_{\eta \text{ child}} M_\eta$. If $S_\xi > 0$, send the fraction M_η/S_ξ of each existing population type to the child η . The mass sent there is $M_\xi M_\eta/S_\xi \leq M_\eta$ by (4.10). Create at that child additional mass $M_\eta - M_\xi M_\eta/S_\xi$ with birth level $k+1$. If $S_\xi = 0$, all these masses are zero. This operation preserves existing particles and fills each child to exactly its prescribed mass. A new particle at a child ending in q_{k+1} receives the entire prefix $(q_1, \dots, q_k, q_{k+1})$ of that child, as well as its new birth level. Thus a particle born below a zero-mass ancestor still has a specified direction history at that ancestor. No particle mass is counted as occupying that ancestor before birth.

After the finite number of levels, the masses of the resulting terminal-path and birth-level types define a finite atomic measure μ . All particles present at ξ continue to terminal descendants of ξ , conserving their mass M_ξ . Conversely, any terminal particle with that prefix and $B \leq k$ was already present at ξ . Particles born later may carry the same prefix, but the condition $B \leq k$ excludes them. This proves (4.12). Summing over prefixes and using (4.11) gives (4.13). For a positive consecutive refinement coefficient, $|q_{j+1} - q_j|_\infty \leq r_j/2$. Hence

$$|q_m - q_s|_\infty \leq \sum_{j=s}^{m-1} r_j/2 = r_s(1 - 2^{-(m-s)}) < r_s,$$

with the same strict upper bound when $s = m$. This proves the locality assertion. $\square$

The particle measure now realizes every vector mass exactly at its level. It lets us prove a pathwise bound and then recover a sum over all cells by integration. We next mark the genuine direction changes along a path. The sum of the norms of descendant vectors may exceed the norm of their sum at the ancestor; this nonnegative difference will pay for the marks.

Mark every root of the path forest. Below a mark, mark the first nodes whose direction θ_{q_s} has distance greater than $\tau/8$ from the direction at that mark, and repeat the rule below each new mark. These intrinsic marks depend on the path forest and the vector masses; they do not depend on the label process at any point x .

Lemma 4.5 (Intrinsic marks and expected switches). *The particle measure satisfies*

$$\sum_{\xi \text{ intrinsically marked}} M_\xi \leq CV, \quad (4.15)$$

$$\int \sum_{s=B}^m E_s(q_s) \, d\mu \leq CV, \quad (4.16)$$

$$\int \sum_{s=B}^m R_s(y) \, d\mu \leq CV. \quad (4.17)$$

Proof. For a marked nonterminal node ξ , let $\mathcal{L}_\xi$ consist of the first subsequent marks on each descending branch, supplemented by terminal nodes on branches with no later mark. This is a finite frontier. Writing $\vartheta_\xi = \theta_{q_k}$ at a level- k node, vector additivity gives

$$\sum_{\eta \in \mathcal{L}_\xi} M_\eta - M_\xi = \frac{1}{2} \sum_{\eta \in \mathcal{L}_\xi} M_\eta |\vartheta_\eta - \vartheta_\xi|^2 \geq \frac{\tau^2}{128} \sum_{\substack{\eta \in \mathcal{L}_\xi \\ \eta \text{ marked}}} M_\eta. \quad (4.18)$$

For clarity, vector additivity on the frontier says $\sum_{\eta \in \mathcal{L}_\xi} M_\eta \vartheta_\eta = M_\xi \vartheta_\xi$. Taking its scalar product with the unit vector ϑ_ξ proves the identity, even if $M_\xi = 0$. For $m > 1$, every marked nonterminal node other than a root occurs positively in its preceding marked frontier and negatively in its own summand. Every terminal node occurs in exactly one frontier. The sum of the left sides is therefore $A_m - A_1 \leq V$. Every nonroot mark occurs in exactly one frontier. Adding the root masses proves (4.15); if $m = 1$, the assertion follows directly from the root mass bound.

By (4.12), integration of the number of intrinsic marks at levels $s \geq B$ is the left side of (4.15). Similarly,

$$\int \sum_{s=B}^m E_s(q_s) \, d\mu = \sum_{s=1}^m \sum_{q \in V_s} a_q E_s(q),$$

so (4.16) follows from (4.1).

For the switch estimate, fix a particle, including its path, birth level, and terminal position y . Run the finite-state law at this fixed point y . Let $X_1 = 1$, and for $s \geq 2$ let $X_s = \mathbf{1}_{\{n_s \neq n_{s-1}\}}$. Thus $\mathbb{E}_y X_s = R_s(y)$ for $1 \leq s \leq m$. Among levels $B, \dots, m$, call a level exceptional if it is intrinsically marked or if $E_s(q_s) > c_*$. Remove those levels and consider a consecutive interval of remaining levels. Throughout this interval, all θ_{q_s} are within $\tau/8$ of the same most recent intrinsic marked direction, even if that mark occurred before birth. Every sampled vertex at y lies within $2r_s$ of q_s by (4.14), and hence its sampled direction is within $\tau/10$ of θ_{q_s} . Any two sampled directions on the interval are therefore at distance at most

$$2(\tau/8 + \tau/10) = 9\tau/20.$$

After one switch on the interval, the new label is within an additional $\tau/20$ of the direction sampled at that switch. Its distance to every subsequently sampled direction is at most $\tau/2 < \tau$, so all later switch probabilities on this interval are zero. The same observation applies after the initial assignment, should it be counted on such an interval. There is consequently at most one counted switch on each remaining interval, regardless of the label entering it.

If there are J exceptional levels, there are at most $J + 1$ such intervals and at most J switches at exceptional levels. Thus every realization obeys

$$\sum_{s=B}^m X_s \leq 1 + 2\#\{s \geq B : s \text{ is intrinsically marked}\} + 2c_*^{-1} \sum_{s=B}^m E_s(q_s).$$

Taking $\mathbb{E}_y$, then integrating over particles, proves (4.17) from the first two bounds and $\mu(\text{all particles}) \leq V$. $\square$

4.4 Comparison of label laws at nearby points

We have bounded the expected switches at the single point y attached to each particle. This does not yet give Proposition 4.3: the local prices are integrated on cells, so their coefficients must be

controlled uniformly there. We also still need the row-derivative cost. The same comparison of nearby rows supplies both missing bounds.

We compare the law at any point of $D(q_l)$ with the law at the terminal position of a particle. The comparison uses the complete particle path, including its levels before birth.

Lemma 4.6 (Spatial comparison). *Fix a particle, a level $1 \leq l \leq m$, and $x \in D(q_l)$. Set*

$$c_s = R_s(y) + E_s(q_s), \quad 1 \leq s \leq m.$$

For $1 \leq k \leq \min(l+1, m)$,

$$R_k(x) + T_k(x) \leq C \sum_{s=1}^k 2^{s-k} c_s. \quad (4.19)$$

The assertion for R_k holds at every x in the specified set, and the assertion for T_k holds almost everywhere there. The constant is uniform over particles, l , and x .

Proof. By (4.14), $|x - y|_\infty < 26r_l$. At a step $s \leq \min(l+1, m)$, any vertex z whose tent is nonzero at x or at y satisfies

$$|z - q_s|_\infty \leq r_s + 26r_l + r_s \leq 54r_s < 200r_s.$$

Off G , a tent with nonzero gradient at x also has positive value there, so the same bound covers every differentiated tent. At grid points the probability comparison still uses the exact tent values; no assertion about a classical row derivative is needed. Thus all the directions involved differ from θ_{q_s} by at most $\sqrt{E_s(q_s)}$.

Fix $s \geq 2$ and an old label n . The function $d \mapsto \sqrt{P(d)} = \min\{1, \tau^{-1}(d - \tau)_+\}$ is τ^{-1} -Lipschitz. For any two involved vertices z, z' , it follows that

$$p_s(n, z) \leq 2p_s(n, z') + 2\tau^{-2}|\Theta_z - \Theta_{z'}|^2 \leq 2p_s(n, z') + 8\tau^{-2}E_s(q_s).$$

Averaging z' with the probabilities $\phi_{z'}(y)$ shows that the largest $p_s(n, z)$ among all involved vertices is bounded by

$$C(\rho_s(y | n) + E_s(q_s)). \quad (4.20)$$

The row formula (4.5) and the tent derivative bounds then give

$$\rho_s(x | n) \leq C(\rho_s(y | n) + E_s(q_s)), \quad (4.21)$$

$$r_s \sum_{n' \in \mathcal{N}} |\nabla_x \pi_s(x; n, n')| \leq C(\rho_s(y | n) + E_s(q_s)) \quad (4.22)$$

almost everywhere in the second assertion. Moreover, the union of the sets of tents nonzero at x and y has uniformly bounded cardinality, and each tent is C/r_s -Lipschitz. Subtracting the two rows therefore yields the pointwise bound

$$\|\pi_s(x; n, \cdot) - \pi_s(y; n, \cdot)\|_{\ell^1} \leq C2^{s-l}(\rho_s(y | n) + E_s(q_s)). \quad (4.23)$$

For $s = 1$, the analogous derivative and difference bounds are C and $C2^{1-l}$, respectively, by (4.4). These bounds are covered by using $c_1 = 1 + E_1(q_1) \geq 1$.

The rows have now been compared for each fixed old label. The old-label distributions themselves also differ between x and y , and must be accounted for before estimating R_k and T_k . Let $d_s(x, y) = \|\zeta_s(x) - \zeta_s(y)\|_{\ell^1}$ and $d_0 = 0$. Multiplication by a stochastic matrix Π contracts the ℓ^1 norm of signed row vectors: for any such vector v ,

$$\|v\Pi\|_1 \leq \sum_n |v(n)| \sum_{n'} \Pi(n, n') = \|v\|_1.$$

The marginal recursion therefore gives

$$\begin{aligned} d_s(x, y) &\leq d_{s-1}(x, y) + \|\zeta_{s-1}(y)(\pi_s(x) - \pi_s(y))\|_{\ell^1} \\ &\leq d_{s-1}(x, y) + C2^{s-l}c_s. \end{aligned}$$

The first step uses the single-row interpretation. At later steps the last inequality follows by averaging (4.23) against the old-label law at y , which averages $\rho_s(y \mid n)$ to $R_s(y)$. Consequently

$$d_j(x, y) \leq C \sum_{s=1}^j 2^{s-l}c_s, \quad j \leq \min(l+1, m). \quad (4.24)$$

This recursion does not assume independence between marginal labels at the two observation points.

For $k \geq 2$, first average the row switch rate and the row derivative cost at x against $\zeta_{k-1}(y)$. Equations (4.21)–(4.22) bound the result by Cc_k . Replacing this old-label distribution by $\zeta_{k-1}(x)$ costs at most $Cd_{k-1}(x, y)$, because row switch rates are at most one and scaled row derivative costs are uniformly bounded. Thus

$$R_k(x) + T_k(x) \leq Cc_k + C \sum_{s=1}^{k-1} 2^{s-l}c_s.$$

Since $k \leq l+1$, we have $2^{s-l} \leq 22^{s-k}$. This proves (4.19) for $k \geq 2$. For $k = 1$ it follows directly from $R_1 = 1$, (4.8), and $c_1 \geq 1$. $\square$

4.5 The packing estimate

Proof of Proposition 4.3. Denote the bracket for a level- l vertex by $\mathcal{C}_l(q)$. All its quantities are nonnegative and measurable; the switch rates are continuous and the derivative costs have specified measurable versions. The particle occupancy identity gives the exact rewriting

$$\sum_{l=1}^m \sum_{q \in V_l} a_q \mathcal{C}_l(q) = \int \sum_{l=B}^m \mathcal{C}_l(q_l) \, d\mu. \quad (4.25)$$

Fix a particle, and continue to write $c_s = R_s(y) + E_s(q_s)$. The comparison in Lemma 4.6 is uniform on each $D(q_l)$, so it applies to the supremum and essential supremum in $\mathcal{C}_l(q_l)$. For $l < m$, it bounds both R_l and R_{l+1} . For $l = m$, the convention $R_{m+1} = 1$ contributes just one additional unit. The derivative terms have the exact geometric convolution

$$\sum_{k=1}^l 2^{k-l} \sum_{s=1}^k 2^{s-k}c_s = \sum_{s=1}^l (l-s+1)2^{s-l}c_s.$$

It follows that

$$\sum_{l=B}^m \mathcal{C}_l(q_l) \leq C \left[1 + \sum_{l=B}^m \sum_{s=1}^{\min(l+1, m)} (l-s+2)2^{s-l}c_s \right]. \quad (4.26)$$

For a fixed $s \geq B$, the coefficient of c_s on the right is bounded by

$$\sum_{j=-1}^{\infty} (j+2)2^{-j} < \infty, \quad j = l-s.$$

Thus the terms at or after birth cost at most $C \sum_{s=B}^m c_s$. Their integral is at most CV by (4.16) and (4.17).

For the terms before birth, use $0 \leq c_s \leq 5$. Put $q = B - s \geq 1$ and $j = l - B \geq 0$. Their total for this particle is at most

$$5 \sum_{q=1}^{\infty} \sum_{j=0}^{\infty} (j+q+2) 2^{-(j+q)} = 5 \sum_{n=1}^{\infty} n(n+2) 2^{-n} < \infty.$$

This bound is independent of B and of the particle, and therefore its integral is at most CV by (4.13). The isolated unit in (4.26) has the same bound. Inserting these estimates into (4.25) proves (4.9). $\square$

5 Signed finite-radius envelopes

The common prices control unfavorable directional components. The direction law controls the endpoints created when a direction changes. We now combine those estimates, keeping the remaining directional projections signed until integration by parts. The essential cancellation is that the derivatives of a probability row sum to zero. Subtracting the potential of the preceding history then leaves only oscillations at smaller radii.

Throughout this section, $g \in C_c^\infty(\mathbb{R}^2)$ is real, $u_t = A_t g$, $V = \int |\nabla g|$, and $r_k = 2^{-k}$ for $1 \leq k \leq m$. Use the prices and label law of Sections 3 and 4. For $1 \leq i \leq j \leq m$, put

$$F_{ij}(x) = \max_{r_j \leq t \leq 2r_i} u_t(x). \quad (5.1)$$

The maximum is attained, since $t \mapsto u_t(x)$ is continuous. These envelopes are compactly supported: all their radii are at most $2r_1 = 1$, so all their averages vanish outside the closed 1-neighborhood of $\text{supp } g$. They are Lipschitz, with common Lipschitz bound $\|\nabla g\|_\infty$.

Outside one null set all the finitely many F_{ij} are differentiable. At any such point x , every radius t attaining $F_{ij}(x)$ satisfies

$$\nabla F_{ij}(x) = \nabla u_t(x). \quad (5.2)$$

Indeed, with this radius held fixed, the differentiable function $F_{ij} - u_t$ is nonnegative everywhere and is zero at x . Its gradient there is zero. This proves the assertion also when there are several attaining radii. The same touching principle underlies the optimizing-radius identity of Hajlasz and Malý [7, Theorem 2 and Remark 3]; here it applies directly to the smooth signed averages. No selected radius is differentiated.

To prove Proposition 1.2, it suffices first to bound the normalized band by an absolute constant independent of m :

$$\int_{\mathbb{R}^2} |\nabla F_{1m}| \, dx \leq C \int_{\mathbb{R}^2} |\nabla g| \, dx. \quad (5.3)$$

Proof of Proposition 1.2. We first bound the normalized band in (5.3). For a label history $\mathbf{n} = (n_1, \dots, n_m) \in \mathcal{N}^m$, a *run* is a maximal interval $[i, j]$ of indices on which the label is constant. Its radius interval is $[r_j, 2r_i]$. Adjacent intervals meet at an endpoint, and their union is $[r_m, 1]$. Thus

$$F_{1m} = \max_{[i,j] \text{ a run}} F_{ij}.$$

At a point where all the envelopes are differentiable, some run attains this maximum, and its gradient agrees with ∇F_{1m} by the same touching argument.

The cone inequality (3.5) states that, for every $n \in \mathcal{N}$ and $z \in \mathbb{R}^2$,

$$|z| \leq C \left[n \cdot z + \sum_{\pm} (-e_{\pm}(n) \cdot z)_+ \right],$$

and the bracket is nonnegative. Apply it to a run attaining F_{1m} , add the nonnegative brackets from the other runs, and average over the history law at x . This gives

$$|\nabla F_{1m}| \leq C \mathbb{E}_x \sum_{[i,j] \text{ a run}} \left[n_i \cdot \nabla F_{ij} + \sum_{\pm} (-e_{\pm}(n_i) \cdot \nabla F_{ij})_+ \right] \quad \text{a.e.} \quad (5.4)$$

There are finitely many histories and runs. In particular, their expectations and all differentiations of their weights below are finite sums.

The unfavorable components and the ends of runs. For a run $[i, j]$, choose any attaining radius t at the point under consideration, and a bin $[r_k, 2r_k]$ containing it, with $i \leq k \leq j$. If $i < k < j$, the interval $[t - r_k/4, t + r_k/4]$ lies inside $[r_j, 2r_i]$. The maximizing property therefore gives $t \in \mathcal{I}_k(x)$. If $k = i$ or $k = j$, the full bin is still contained in the run interval, so t maximizes on that bin and belongs to $\mathcal{J}_k(x)$. Touching, including at tied radii, identifies the gradients.

Write

$$B_k(x) = \sup_{t \in \mathcal{J}_k(x)} |\nabla u_t(x)|, \quad A_{k,n}(x) = \sup_{t \in \mathcal{I}_k(x)} \sum_{\pm} (-e_{\pm}(n) \cdot \nabla u_t(x))_+.$$

A bin is an end bin of its run only if a run starts at k or ends at k . These events have probabilities R_k and R_{k+1} , respectively, with the sentinel conventions $R_1 = R_{m+1} = 1$. Also each bin belongs to exactly one run. Since each of the two negative projections is bounded by the gradient norm, we obtain

$$\mathbb{E}_x \sum_{[i,j] \text{ a run}} \sum_{\pm} (-e_{\pm}(n_i) \cdot \nabla F_{ij})_+ \leq \sum_{k=1}^m \sum_{n \in \mathcal{N}} \mathbb{P}_x(n_k = n) A_{k,n}(x) + 2 \sum_{k=1}^m (R_k + R_{k+1}) B_k(x). \quad (5.5)$$

No measurable choice of the attaining radius is needed: this is a pointwise bound by the measurable suprema on the right.

Here are the spatial sums in this estimate. For a level- k cell $Q = Q_p$ and $q \in \mathcal{R}(Q)$, every $x \in Q$ satisfies $|x - q|_{\infty} \leq 21r_k$. Thus $Q \subset D(q)$, and Q also lies in the $50r_k$ neighborhood used in Lemma 4.2. If (q, n) is bad and $E_k(q) \leq c_*$, that lemma says $\mathbb{P}_x(n_k = n) = 0$ throughout Q . Every remaining bad-root charge is bounded by $a_q \leq c_*^{-1} a_q E_k(q)$. Take the supremum of each label probability on Q and apply the directional part of Proposition 3.3. Its price charges sum by (3.15); its root charges sum by (4.1). The finite palette and the 41^2 cells having a given vertex in their root neighborhood contribute only absolute factors.

For the last sum in (5.5), the endpoint estimate in Proposition 3.3 gives, cell by cell,

$$\int_Q (R_k + R_{k+1}) B_k \leq C \sup_Q (R_k + R_{k+1}) \left(D_Q + \sum_{q \in \mathcal{R}(Q)} a_q \right).$$

The price part sums because $R_k + R_{k+1} \leq 2$. For each root part, enlarge Q to $D(q)$ and use the switch part of (4.9), again with bounded cell multiplicity. We have proved

$$\int_{\mathbb{R}^2} \mathbb{E}_x \sum_{[i,j] \text{ a run}} \sum_{\pm} (-e_{\pm}(n_i) \cdot \nabla F_{ij})_+ \, dx \leq CV. \quad (5.6)$$

Baselines and weighted integration by parts. It remains to integrate the signed projections in (5.4). Subtract u_{r_i} from the envelope of every run starting at i . Since r_i is a forced endpoint in $\mathcal{J}_i(x)$, the same endpoint and packing estimates give

$$\int \mathbb{E}_x \sum_{[i,j] \text{ a run}} |\nabla u_{r_i}| = \sum_{i=1}^m \int R_i |\nabla u_{r_i}| \leq \sum_{i=1}^m \int (R_i + R_{i+1}) B_i \leq CV. \quad (5.7)$$

For a fixed history define its vector potential by

$$\Phi_{\mathbf{n}}(x) = \sum_{[i,j] \text{ a run}} n_i (F_{ij}(x) - u_{r_i}(x)). \quad (5.8)$$

Its run partition and labels are fixed; only its scalar functions depend on x . It is a compactly supported Lipschitz vector field, and its divergence is the sum of the signed projections after baseline subtraction.

Use the auxiliary state $n_0 = *$ to write the initial row as $\pi_1(x; *, n_1)$. The probability of the history is

$$w_{\mathbf{n}}(x) = \prod_{s=1}^m \pi_s(x; n_{s-1}, n_s).$$

Every factor is Lipschitz and takes values in $[0, 1]$. The weak product rule and compact support of $w_{\mathbf{n}} \Phi_{\mathbf{n}}$ therefore give

$$\sum_{\mathbf{n}} \int w_{\mathbf{n}} \operatorname{div} \Phi_{\mathbf{n}} \, dx = - \sum_{\mathbf{n}} \int \Phi_{\mathbf{n}} \cdot \nabla w_{\mathbf{n}} \, dx. \quad (5.9)$$

Indeed the integral of the divergence of a compactly supported Lipschitz vector field is zero, as follows by testing against a smooth function equal to one on its support. We must exploit the sign of the right side before estimating its size.

Cancellation after a fixed prefix. Differentiate the product defining $w_{\mathbf{n}}$ directly. For its term at step k , fix the prefix $\mathbf{a} = (n_1, \dots, n_{k-1})$ and set

$$W_{\mathbf{a}} = \prod_{s < k} \pi_s(x; n_{s-1}, n_s), \quad J_{k, \mathbf{n}} = \prod_{s > k} \pi_s(x; n_{s-1}, n_s).$$

Empty products equal one. The corresponding term of $\nabla w_{\mathbf{n}}$ is $W_{\mathbf{a}} \nabla \pi_k(x; n_{k-1}, n_k) J_{k, \mathbf{n}}$. This formula remains valid when some transition entries vanish; it involves no division by them.

For each fixed prefix and n_k , summing the future product gives

$$\sum_{n_{k+1}, \dots, n_m} J_{k, \mathbf{n}} = 1. \quad (5.10)$$

To see this, first sum in n_m using the normalization of the last row, and repeat backward to n_{k+1} . Also, wherever the rows are differentiable,

$$\sum_{n_k \in \mathcal{N}} \nabla \pi_k(x; n_{k-1}, n_k) = 0, \quad (5.11)$$

by differentiating their finite sum, which is the constant one.

Let $\Psi_{\mathbf{a}}$ be the potential of this prefix with its last run closed at $k-1$: explicitly, use the sum (5.8) for the runs of $(n_1, \dots, n_{k-1})$. For $k=1$ set $\Psi_{\varnothing} = 0$. This potential is independent of n_k and of every future label. Equations (5.10)–(5.11) therefore imply the exact identity

$$\begin{aligned} & \sum_{n_k, \dots, n_m} J_{k, \mathbf{n}} \Phi_{\mathbf{n}} \cdot \nabla \pi_k(x; n_{k-1}, n_k) \\ &= \sum_{n_k, \dots, n_m} J_{k, \mathbf{n}} (\Phi_{\mathbf{n}} - \Psi_{\mathbf{a}}) \cdot \nabla \pi_k(x; n_{k-1}, n_k). \end{aligned} \quad (5.12)$$

The subtraction precedes every absolute value in the next estimate.

We claim that each continuation of the prefix satisfies

$$|\Phi_{\mathbf{n}}(x) - \Psi_{\mathbf{a}}(x)| \leq \sum_{l=k}^m v_l(x), \quad v_l = \sup_{0 < t, t' \leq 4r_l} |u_t - u_{t'}|. \quad (5.13)$$

All already completed runs cancel. If the last prefix run $[i, k-1]$ continues to $j \geq k$, its baseline cancels as well, leaving $n_i(F_{ij} - F_{i, k-1})$. The old radius interval contains r_{k-1} , and the extension adds only radii below that number. Consequently

$$0 \leq F_{ij} - F_{i, k-1} \leq \sup_{0 < t \leq r_{k-1}} (u_t - u_{r_{k-1}})_+ \leq v_k, \quad r_{k-1} = 2r_k.$$

Every new run $[l, j]$, beginning at $l \geq k$, contributes a vector of norm $F_{lj} - u_{r_l} \leq v_l$, because its envelope includes the baseline and all its radii are at most $2r_l$. If the prefix run continues, the new runs start strictly after k ; if there is a switch at k , the closed prefix itself is unchanged. The empty-prefix case consists entirely of new runs. Since their starting indices are distinct, these observations prove (5.13). In particular, the size of an old large-radius envelope never enters the bound.

Now take absolute values in (5.12) and sum over prefixes. The future products are nonnegative and sum to one. The sum of the norms of the step- k product-rule terms is exactly

$$\begin{aligned} & \sum_{\mathbf{a}} W_{\mathbf{a}} \sum_{n_k} |\nabla \pi_k(x; n_{k-1}, n_k)| \sum_{n_{k+1}, \dots, n_m} J_{k, \mathbf{n}} \\ &= \sum_{n \in \mathcal{N}} \zeta_{k-1}(x; n) \sum_{n'} |\nabla \pi_k(x; n, n')| = \frac{T_k(x)}{r_k} \quad (k \geq 2). \end{aligned}$$

For $k=1$ the same identity uses the single initial row, exactly as in (4.7). Combining these identities with the product expansion and (5.9) yields

$$\left| \sum_{\mathbf{n}} \int w_{\mathbf{n}} \operatorname{div} \Phi_{\mathbf{n}} \right| \leq \sum_{1 \leq k \leq l \leq m} \int \frac{T_k(x)}{r_k} v_l(x) \, dx. \quad (5.14)$$

Paying for the smaller-radius oscillations. On a level- l cell, the value estimate in Proposition 3.3 gives

$$\int_Q v_l \leq Cr_l \left(D_Q + \sum_{q \in \mathcal{R}(Q)} a_q \right).$$

Multiply this by $r_k^{-1} \operatorname{esssup}_Q T_k$. For the price terms, $T_k \leq C$ and

$$\sum_{k=1}^l \frac{r_l}{r_k} = \sum_{k=1}^l 2^{k-l} \leq 2.$$

Their full sum is therefore at most $C \sum_{l,Q} D_Q \leq CV$. For the root terms use $Q \subset D(q)$ and reorder the finite nonnegative sums. They are at most an absolute multiple of

$$\sum_{l=1}^m \sum_{q \in V_l} a_q \sum_{k=1}^l 2^{k-l} \operatorname{ess\,sup}_{D(q)} T_k \leq CV$$

by (4.9). Thus (5.14) is at most CV . Together with (5.7), this bounds the integral of the signed projections in (5.4) from above by CV . Equation (5.6) pays for the remaining terms and proves (5.3).

For $b/a = 2^m$ with $m \geq 1$, apply the normalized result to $g_b(x) = g(bx)$. Its normalized envelope is $S_{a,b}g(bx)$, and both gradient integrals acquire the same factor b^{-1} under the change of variables. This proves the stated bound for all bands in the proposition. Finally, when $a = b$, convolution gives $\nabla A_a g = A_a(\nabla g)$, and Tonelli's theorem yields $\int |\nabla A_a g| \leq \int |\nabla g|$. $\square$

6 Sobolev input and absolute continuity

The signed estimate now gives the endpoint theorem in two steps. First, approximation passes it to Sobolev inputs and yields a finite distributional derivative measure for the maximal function. Second, local mean subtraction shows that this measure gives no mass to a Lebesgue-null set. The second step is needed to obtain a weak gradient in L^1 , and it is the reason the finite-band estimate was proved for signed inputs.

For real $g \in W^{1,1}(\mathbb{R}^2)$ and $0 < a \leq b < \infty$, continue to write

$$S_{a,b}g(x) = \sup_{a \leq t \leq b} A_t g(x).$$

The disk integrals define these functions independently of the representative of g . The positive lower radius makes the supremum finite at every x , since $|A_t g(x)| \leq \|g\|_1/(\pi a^2)$. The elementary analytic facts used below, including the exact Euclidean variation formula, are proved in Appendix A.

Lemma 6.1 (Sobolev finite bands). *For every real $g \in W^{1,1}(\mathbb{R}^2)$ and $0 < a \leq b < \infty$, $S_{a,b}g$ is globally Lipschitz and*

$$\operatorname{Lip}(S_{a,b}g) \leq \frac{\|\nabla g\|_1}{\pi a^2}. \quad (6.1)$$

If b/a is a nonnegative integer power of two, then

$$\int_{\mathbb{R}^2} |\nabla S_{a,b}g| \, dx \leq C \|\nabla g\|_1, \quad (6.2)$$

where C is absolute and independent of the band. This includes $a = b$, when the envelope is one average.

Proof. The translation estimate of Lemma A.1 gives, for $t \geq a$ and $z \in \mathbb{R}^2$,

$$\begin{aligned} |A_t g(x+z) - A_t g(x)| &\leq \frac{1}{\pi t^2} \int_{B(x,t)} |g(y+z) - g(y)| \, dy \\ &\leq \frac{|z|}{\pi a^2} \|\nabla g\|_1. \end{aligned}$$

Taking suprema over the same radius interval on both sides proves (6.1). The same comparison gives, for any two integrable real inputs,

$$\|S_{a,b}g - S_{a,b}\tilde{g}\|_\infty \leq \frac{\|g - \tilde{g}\|_1}{\pi a^2}. \quad (6.3)$$

Here the inequality $|\sup f_t - \sup h_t| \leq \sup |f_t - h_t|$ is applied to bounded families of real numbers.

Choose real $g_j \in C_c^\infty$ converging to g in $W^{1,1}$. At the fixed lower radius a , (6.3) gives uniform convergence of the band envelopes. If b/a is a power of two greater than one, Proposition 1.2 therefore passes to the limit by the local variation inequality in Lemma A.4:

$$\int |\nabla S_{a,b}g| \leq \liminf_j \int |\nabla S_{a,b}g_j| \leq C \lim_j \|\nabla g_j\|_1 = C\|\nabla g\|_1.$$

The limit is Lipschitz by (6.1), so its weak and almost-everywhere classical gradients agree. When $a = b$, testing the convolution against smooth functions gives $\nabla A_a g = A_a(\nabla g)$ distributionally; the integral of its norm is at most $\|\nabla g\|_1$ by Tonelli's theorem. $\square$

Proof of Theorem 1.1. At this point take $g = |f|$. Lemma A.1 shows that $g \in W^{1,1}(\mathbb{R}^2)$ and

$$V_g := \|\nabla g\|_1 \leq \|\nabla f\|_1.$$

The planar Sobolev inequality in Lemma A.2 gives $g \in L^2(\mathbb{R}^2)$. The strong L^2 maximal inequality [10, (1.2)] then yields $Mf = Mg \in L^2(\mathbb{R}^2)$, hence local integrability. Define

$$G_N = S_{2^{-N}, 2^N} g, \quad N = 1, 2, \dots$$

Because $g \geq 0$, these nonnegative functions increase to Mg and are bounded above by it. Dominated convergence on every bounded set implies $G_N \rightarrow Mg$ in L^1_{loc} .

A finite derivative measure. Lemma 6.1 bounds $\int |\nabla G_N|$ by CV_g uniformly in N . In particular, for every $\varphi \in C_c^1(\mathbb{R}^2; \mathbb{R}^2)$,

$$\left| - \int Mg \operatorname{div} \varphi \right| = \lim_N \left| \int \nabla G_N \cdot \varphi \right| \leq CV_g \|\varphi\|_\infty.$$

The vector-measure statement of Lemma A.4 supplies a finite vector Radon measure $H = D(Mg)$, with

$$|H|(\mathbb{R}^2) \leq CV_g. \quad (6.4)$$

It also gives the local estimate

$$|H|(O) \leq \liminf_{N \rightarrow \infty} \int_O |\nabla G_N| \, dx \quad (O \subset \mathbb{R}^2 \text{ open}). \quad (6.5)$$

These conclusions control total variation. We next show that H is absolutely continuous.

Lower-radius Lipschitz truncations appear in Hajlasz and Malý [7, Lemma 8] and Kurka [11, Lemma 7.1]; local/nonlocal splitting with subtraction of the input mean appears in Saari [14, Section 3.1]. Here the signed band estimate and Poincaré's inequality make the localization cost depend only on the nearby input gradient. We give that argument in full.

Small and large radii near a null set. Fix a compact Lebesgue-null set K . The assertion is immediate if K is empty, so assume it is nonempty. Fix also a dyadic $t = 2^{-L}$ with $L \geq 1$, and an open set

$$K \subset O \subset \{x : \operatorname{dist}(x, K) < t\}.$$

For every $N \geq L$, splitting the band at t gives

$$G_N = \max\{S_{2^{-N}, t}g, S_{t, 2^N}g\}.$$

Both functions are Lipschitz. The maximum rule, including the equality set, in Lemma A.1 gives

$$|\nabla G_N| \leq |\nabla S_{2^{-N},t}g| + |\nabla S_{t,2^N}g| \quad \text{a.e.}$$

For the second term, (6.1) bounds its integral over O by

$$\int_O |\nabla S_{t,2^N}g| \leq \frac{|O|}{\pi t^2} V_g. \quad (6.6)$$

For the first term, partition the plane into half-open grid squares Q of side t , and retain those meeting O . There are finitely many, since O is contained in a bounded neighborhood of K . Choose the cutoff of Lemma A.3, equal to one on $3Q$ and supported in $5Q$, and put

$$c_Q = \frac{1}{|5Q|} \int_{5Q} g, \quad g_Q = \chi_Q(g - c_Q).$$

The product rule and Poincaré's inequality give

$$\|\nabla g_Q\|_1 \leq C \int_{5Q} |\nabla g|. \quad (6.7)$$

Every disk of radius $s \leq t$ centered in Q lies in $3Q$. Consequently

$$A_s g_Q(x) = A_s g(x) - c_Q \quad (x \in Q, 0 < s \leq t), \quad S_{2^{-N},t} g_Q = S_{2^{-N},t} g - c_Q \quad \text{on } Q.$$

The two envelopes have the same gradient almost everywhere on Q . Since g_Q can have either sign, we use precisely the signed form of Lemma 6.1. The ratio of its endpoints is 2^{N-L} , including one when $N = L$. It follows that

$$\begin{aligned} \int_O |\nabla S_{2^{-N},t}g| &\leq \sum_{Q \cap O \neq \emptyset} \int_Q |\nabla S_{2^{-N},t}g_Q| \\ &\leq C \sum_{Q \cap O \neq \emptyset} \|\nabla g_Q\|_1 \leq C \int_{\{\text{dist}(x,K) < 6t\}} |\nabla g(x)| \, dx. \end{aligned}$$

The last step uses the bounded overlap of the squares $5Q$ and their containment in the $6t$ -neighborhood of K , both established in Lemma A.3. Grid boundaries are null and do not affect these gradient integrals.

Combining this estimate with (6.6) and then (6.5) yields

$$|H|(O) \leq C \int_{\{\text{dist}(x,K) < 6t\}} |\nabla g(x)| \, dx + Ct^{-2}|O|V_g. \quad (6.8)$$

The constants are independent of N , O , and t .

Removing the singular part. For the fixed t , compactness and Lebesgue nullity of K allow the open neighborhood O above to have arbitrarily small area. Since $|H|(K) \leq |H|(O)$, (6.8) gives

$$|H|(K) \leq C \int_{\{\text{dist}(x,K) < 6t\}} |\nabla g(x)| \, dx.$$

Only now let t decrease to zero through dyadic values. The neighborhoods decrease to the compact null set K , and $|\nabla g|$ is integrable; the right side tends to zero. Thus $|H|(K) = 0$ for every compact

Lebesgue-null set. Inner regularity of the finite Radon measure $|H|$ implies that it vanishes on every Borel null set, and hence on every Lebesgue-null set by containment in a Borel null set.

The Radon–Nikodym conclusion of Lemma A.4 now gives $H = G \, dx$ with $G \in L^1(\mathbb{R}^2; \mathbb{R}^2)$ and

$$\int |G| = |H|(\mathbb{R}^2) \leq CV_g \leq C \int |\nabla f|.$$

For every smooth compactly supported vector field φ , the defining identity of H reads $-\int Mf \operatorname{div} \varphi = \int G \cdot \varphi$. Thus G is the weak gradient of Mf . Since Mf is locally integrable, this proves $Mf \in W_{\text{loc}}^{1,1}$ and (1.1), with a globally integrable Euclidean gradient. $\square$

Remark 6.2. The direct absolute-continuity argument uses the signed band estimate, approximation, and square Poincaré localization. After (6.4), one could alternatively apply Lahti and Weigt’s centered regularity theorem [12, Theorem 1.2] to $g = |f|$ on $\mathbb{R}^2$. The input is globally integrable and has finite variation, and $Mg \in BV_{\text{loc}}$ has already been proved. Its input jump and Cantor derivative measures vanish, so the theorem’s additional condition on the Cantor part is vacuous; its conclusion gives local Sobolev regularity. This optional route is used only after the variation bound and supplies no part of the finite-band estimate.

Corollary 6.3 (Bounded-variation inputs). *Let $BV(\mathbb{R}^2)$ denote the globally integrable functions whose distributional derivatives have finite total variation. For every real $f \in BV(\mathbb{R}^2)$,*

$$Mf \in L^2(\mathbb{R}^2) \cap BV_{\text{loc}}(\mathbb{R}^2), \quad |D(Mf)|(\mathbb{R}^2) \leq C|Df|(\mathbb{R}^2),$$

with the constant from Theorem 1.1. Writing $P(F) = |D\mathbf{1}_F|(\mathbb{R}^2)$ for perimeter, every measurable set E with $|E| < \infty$ and $P(E) < \infty$ satisfies

$$\int_0^1 P(\{M\mathbf{1}_E > t\}) \, dt \leq CP(E).$$

These superlevel sets have finite measure for every $t > 0$ and finite perimeter for almost every $t > 0$.

Proof. Let $f_\varepsilon = f * \rho_\varepsilon$ for a smooth nonnegative compactly supported probability approximate identity. Then $f_\varepsilon \in W^{1,1}(\mathbb{R}^2)$, $f_\varepsilon \rightarrow f$ in L^1 , and

$$\nabla f_\varepsilon = (Df) * \rho_\varepsilon, \quad \|\nabla f_\varepsilon\|_1 \leq |Df|(\mathbb{R}^2).$$

The variation formula and lower semicontinuity show that these gradient norms tend to $|Df|(\mathbb{R}^2)$, so the approximation is strict in BV . Lemma A.2 bounds $\|f_\varepsilon\|_2$ uniformly; an almost-everywhere convergent subsequence and Fatou’s lemma give $f \in L^2$. The convolution approximate identity therefore also converges to f in L^2 . The pointwise inequality $|Mf_\varepsilon - Mf| \leq M(f_\varepsilon - f)$ and the strong L^2 maximal inequality yield $Mf \in L^2$ and $Mf_\varepsilon \rightarrow Mf$ in L^2 , hence locally in L^1 . Theorem 1.1 bounds the global weak-gradient norm of Mf_ε by $C|Df|(\mathbb{R}^2)$. Lemma A.4 now gives the asserted finite derivative measure and local BV regularity.

For $u = M\mathbf{1}_E$ we have $0 \leq u \leq 1$. Apply the BV coarea formula [5] on $B(0, R)$ and then use monotone exhaustion as $R \rightarrow \infty$ to obtain

$$\int_0^1 P(\{u > t\}) \, dt = |Du|(\mathbb{R}^2) \leq CP(E).$$

Also $|\{u > t\}| \leq t^{-2}\|u\|_2^2 < \infty$ for every $t > 0$. $\square$

This conclusion does not assert global integrability of Mf or absolute continuity of its derivative measure for arbitrary BV input.

A Analytic facts used in the limiting argument

We record the approximation, localization, and measure arguments used in Section 6. All gradient norms are Euclidean. The Sobolev space is inhomogeneous: the function and its distributional first derivatives belong to $L^1(\mathbb{R}^2)$. General Sobolev calculus is treated in [5]; the exact chain, product, and Lipschitz facts used here also appear in [9, Chapter 3, Propositions 3.21–3.22 and Theorems 3.38–3.39]. We give the needed calculations, including the variation formula that preserves the Euclidean norm.

A.1 Approximation, translations, and equality sets

Lemma A.1. *For real $u \in W^{1,1}(\mathbb{R}^2)$ there are real $u_j \in C_c^\infty(\mathbb{R}^2)$ converging to u in $W^{1,1}$, and*

$$\|u(\cdot + z) - u\|_1 \leq |z| \|\nabla u\|_1 \quad (z \in \mathbb{R}^2). \quad (\text{A.1})$$

Moreover $|u| \in W^{1,1}(\mathbb{R}^2)$ with $|\nabla|u|| \leq |\nabla u|$ almost everywhere. If χ is smooth and compactly supported, then $\chi u \in W^{1,1}$ and $\nabla(\chi u) = \chi \nabla u + u \nabla \chi$. For locally Lipschitz real functions v, w , their maximum is locally Lipschitz and satisfies, almost everywhere,

$$\nabla \max(v, w) = \begin{cases} \nabla v, & v > w, \\ \nabla w, & w > v, \\ \nabla v = \nabla w, & v = w. \end{cases} \quad (\text{A.2})$$

Products of locally Lipschitz functions satisfy the weak product rule. Their weak derivatives agree almost everywhere with their classical derivatives.

Proof. The smooth-cutoff product rule follows directly by testing the distributional derivative of u against $\chi \varphi$. Choose χ_R equal to one on $B(0, R)$, zero outside $B(0, 2R)$, with $0 \leq \chi_R \leq 1$ and $|\nabla \chi_R| \leq C/R$. Then

$$\|\chi_R u - u\|_1 \rightarrow 0, \quad \|\nabla(\chi_R u) - \nabla u\|_1 \leq \int_{|x| > R} |\nabla u| + \frac{C}{R} \|u\|_1 \rightarrow 0.$$

Convolving $\chi_R u$ with a smooth compactly supported approximate identity commutes with its distributional derivatives. The convolutions converge in L^1 , for both the function and its gradient. This last fact follows from continuity of translations in L^1 and the approximate-identity formula; translation continuity follows first for continuous compactly supported functions by uniform continuity, and then for L^1 functions by approximation. A diagonal choice of R and convolution radius proves the asserted C_c^∞ density.

For each smooth approximant, the fundamental theorem of calculus along a segment gives

$$|u_j(x + z) - u_j(x)| \leq |z| \int_0^1 |\nabla u_j(x + sz)| \, ds.$$

Integrating in x and using translation invariance proves the translation bound for u_j . Convergence in $W^{1,1}$ proves (A.1) for u .

Here is also a direct smooth-chain argument for the absolute value. For a continuously differentiable function F with bounded derivative and $F(0) = 0$, smooth approximation gives

$$\nabla F(u) = F'(u) \nabla u.$$

Indeed $F(u_j) \rightarrow F(u)$ in L^1 . Along a subsequence $u_j \rightarrow u$ almost everywhere, and

$$F'(u_j)\nabla u_j - F'(u)\nabla u = F'(u_j)(\nabla u_j - \nabla u) + (F'(u_j) - F'(u))\nabla u$$

tends to zero in L^1 , by boundedness and dominated convergence. Passing the smooth identity to distributions proves the formula. Apply it to $F_\varepsilon(s) = \sqrt{s^2 + \varepsilon^2} - \varepsilon$. These functions satisfy $0 \leq F_\varepsilon(u) \leq |u|$ and converge to $|u|$ in L^1 . Their derivatives converge pointwise to $\text{sgn}(u)$, with value zero at $u = 0$, and are bounded by one. Passing to the limit gives $\nabla|u| = \text{sgn}(u)\nabla u$ and the claimed norm inequality.

A locally Lipschitz function restricts to a Lipschitz, hence absolutely continuous, function on each coordinate line in a rectangle. One-dimensional integration by parts and Fubini's theorem give its weak partial derivatives. Rademacher's theorem identifies these with its classical gradient almost everywhere [9, Chapter 3, Theorems 3.38–3.39]. The usual product rule at joint differentiability points, or equivalently the one-dimensional rule on those lines, proves the weak Lipschitz product rule.

For the equality set in (A.2), put $a = v - w$. At almost every point of $\{a = 0\}$ the set has density one and a is differentiable. Its derivative there is zero: a nonzero derivative would give a cone of fixed positive relative area on which a is nonzero in every sufficiently small ball, contrary to density one. Thus $\nabla v = \nabla w$ almost everywhere on their equality set. At such a point, $a(x + h) = o(|h|)$ implies $a_+(x + h) = o(|h|)$, so a_+ also has derivative zero. On the open sets $a > 0$ and $a < 0$, its gradient is respectively ∇a and zero. Since $\max(v, w) = w + a_+$, this proves (A.2). $\square$

Lemma A.2 (Planar Sobolev embedding). *For every real $u \in W^{1,1}(\mathbb{R}^2)$,*

$$\|u\|_2^2 \leq \|\partial_1 u\|_1 \|\partial_2 u\|_1, \quad \text{in particular } \|u\|_2 \leq \|\nabla u\|_1.$$

Proof. For $u \in C_c^\infty$, set

$$a(y) = \int_{\mathbb{R}} |\partial_1 u(x, y)| \, dx, \quad b(x) = \int_{\mathbb{R}} |\partial_2 u(x, y)| \, dy.$$

The fundamental theorem of calculus on the two coordinate lines gives $|u(x, y)| \leq a(y)$ and $|u(x, y)| \leq b(x)$. Thus $|u(x, y)|^2 \leq a(y)b(x)$; integrating proves the first inequality. For general u , approximate as in Lemma A.1. Applying the smooth inequality to differences shows that the approximants are Cauchy in L^2 . Their L^2 limit agrees almost everywhere with their L^1 limit u , by passage to a common almost-everywhere convergent subsequence. Pass to the limit in the inequality. Each coordinate derivative has L^1 norm at most $\|\nabla u\|_1$, giving the last assertion. $\square$

A.2 Square localization

Lemma A.3. *For a square Q of side length $t > 0$ and $u \in W^{1,1}(\mathbb{R}^2)$, write $u_Q = |Q|^{-1} \int_Q u$. Then*

$$\int_Q |u - u_Q| \leq Ct \int_Q |\nabla u|. \quad (\text{A.3})$$

For every grid square Q of side t , there is a smooth cutoff χ_Q equal to one on the closed concentric dilation $3Q$, supported in the interior of $5Q$, with $0 \leq \chi_Q \leq 1$ and $|\nabla \chi_Q| \leq C/t$. With $c_Q = u_{5Q}$, it satisfies

$$\|\nabla[\chi_Q(u - c_Q)]\|_1 \leq C \int_{5Q} |\nabla u|. \quad (\text{A.4})$$

Every disk of radius at most t centered in Q lies in $3Q$. The dilations $5Q$ of grid squares have bounded overlap. If the squares under consideration meet $O \subset \{x : \text{dist}(x, K) < t\}$, all their dilations $5Q$ lie in $\{x : \text{dist}(x, K) < 6t\}$. Constants are independent of Q, t, u, O, K .

Proof. By rotation and translation it suffices to take $Q = (0, t)^2$. First suppose u is smooth. Jensen's inequality gives

$$\int_Q |u - u_Q| \leq t^{-2} \int_Q \int_Q |u(x) - u(y)| \, dy \, dx.$$

Join $x = (x_1, x_2)$ to (y_1, x_2) and then to $y = (y_1, y_2)$. For the first segment, with x_2 fixed, integrating its derivative bound in x_1, y_1 gives at most

$$t^2 \int_0^t |\partial_1 u(s, x_2)| \, ds.$$

Indeed each s lies between x_1 and y_1 for a set of pairs of area at most t^2 . Integration in x_2 and y_2 therefore gives $t^3 \int_Q |\partial_1 u|$. The second segment similarly costs at most $t^3 \int_Q |\partial_2 u|$. After division by t^2 , the bound $|\partial_1 u| + |\partial_2 u| \leq \sqrt{2} |\nabla u|$ proves (A.3). Global smooth approximation proves it for the stated Sobolev input, since both the functions and their means converge on Q .

Rescale a fixed smooth cutoff to obtain χ_Q . The product rule and (A.3), now on $5Q$, yield

$$\|\nabla[\chi_Q(u - c_Q)]\|_1 \leq \int_{5Q} |\nabla u| + \frac{C}{t} \int_{5Q} |u - c_Q| \leq C \int_{5Q} |\nabla u|.$$

In each coordinate the disk containment follows from $t/2 + t = 3t/2$. A point lies in only a fixed number of the dilated squares, since their centers form a grid of spacing t . Finally, if $x \in Q \cap O$ and $y \in 5Q$, then $|y - x| \leq 3\sqrt{2}t$. Hence $\text{dist}(y, K) < (1 + 3\sqrt{2})t < 6t$, as claimed. $\square$

A.3 Vector variation and local lower semicontinuity

For $u \in L^1_{\text{loc}}(\mathbb{R}^2)$ and open $O \subset \mathbb{R}^2$, define its distributional variation on O by

$$\mathcal{V}(u; O) = \sup \left\{ - \int u \, \text{div} \, \varphi : \varphi \in C_c^1(O; \mathbb{R}^2), |\varphi(x)| \leq 1 \right\}. \quad (\text{A.5})$$

The test-field norm is Euclidean. The class is closed under negation, so the same supremum results if the tested integral is replaced by its absolute value.

Lemma A.4. *Suppose $u_j \in W^{1,1}_{\text{loc}}(\mathbb{R}^2)$ converge locally in L^1 to u , and $\sup_j \int_{\mathbb{R}^2} |\nabla u_j| \leq M < \infty$. Then the distributional gradient of u is a finite vector Radon measure H , with*

$$|H|(\mathbb{R}^2) \leq M, \quad |H|(O) = \mathcal{V}(u; O) \leq \liminf_j \int_O |\nabla u_j| \quad \text{for every open } O. \quad (\text{A.6})$$

If $|H|(K) = 0$ for every compact Lebesgue-null set K , then $H = G \, dx$ for some $G \in L^1(\mathbb{R}^2; \mathbb{R}^2)$ and $\|G\|_1 = |H|(\mathbb{R}^2)$. In that case $u \in W^{1,1}_{\text{loc}}$, with weak gradient G .

Proof. For every smooth compactly supported vector field φ , local L^1 convergence implies

$$\left| - \int u \, \text{div} \, \varphi \right| = \lim_j \left| \int \nabla u_j \cdot \varphi \right| \leq M \|\varphi\|_\infty.$$

Smooth compactly supported fields are uniformly dense in $C_0(\mathbb{R}^2; \mathbb{R}^2)$, the continuous fields vanishing at infinity: first cut off outside a large compact set and then convolve. Thus the functional extends uniquely to this space with norm at most M . Apply the scalar Riesz representation theorem to its two coordinates to obtain finite signed Radon measures H_1, H_2 . Their vector $H = (H_1, H_2)$ represents the functional and therefore the distributional gradient of u . The Riesz and Radon–Nikodym theorems used here are recorded in [15, Chapter 1, Theorem 5.14 and Equation (4.17)].

We spell out why the representation retains the Euclidean total variation rather than the sum of coordinate variations. Put $\lambda = |H_1| + |H_2|$ and write $H = q\lambda$ using the scalar Radon–Nikodym theorem. The coordinate densities of q have absolute value at most one. For a Borel set E , the definition of vector total variation by finite Borel partitions gives

$$|H|(E) = \int_E |q| \, d\lambda.$$

The upper bound follows from the triangle inequality on each part. For the reverse bound, partition the bounded range of q into finitely many sets of diameter at most ε , and choose a vector q_s in each. On the corresponding partition E_s of E ,

$$\sum_s |H(E_s)| \geq \sum_s |q_s| \lambda(E_s) - \varepsilon \lambda(E) \geq \int_E |q| \, d\lambda - 2\varepsilon \lambda(E).$$

Let $\varepsilon \downarrow 0$. Set $\mu = |H|$ and $\sigma = q/|q|$ where $q \neq 0$, with value zero elsewhere; then $H = \sigma\mu$ and $|\sigma| = 1$ almost everywhere for μ .

For every open O , this representation gives

$$\mu(O) = \sup \left\{ \int \varphi \cdot dH : \varphi \in C_c(O; \mathbb{R}^2), |\varphi| \leq 1 \right\}. \quad (\text{A.7})$$

One inequality is immediate. For the other, approximate σ on O in $L^1(\mu)$ by simple vector functions with values in the closed unit ball. Inner regularity permits replacement of their finitely many disjoint measurable sets by disjoint compact subsets of O , losing arbitrarily little μ -mass. Choose continuous cutoff functions equal to one on these compact sets, with mutually disjoint compact supports in O , and multiply by the corresponding vector values. Their sum has norm at most one and approximates σ in $L^1(\mu)$, with arbitrarily small error. This proves the reverse inequality. These are the usual polar and open-set variation formulas; see also [15, Chapter 2, Equations (2.1)–(2.4)].

Each continuous compactly supported test field in (A.7) can be extended by zero and convolved with a nonnegative smooth probability kernel of sufficiently small support. The resulting smooth fields remain supported in O , have norm at most one, and converge uniformly. Consequently the same supremum is obtained with C_c^1 test fields. The distributional identity identifies it with $\mathcal{V}(u; O)$ in (A.5). Taking $O = \mathbb{R}^2$ and using the functional norm bound gives $|H|(\mathbb{R}^2) \leq M$.

For each fixed admissible field supported in O ,

$$-\int u \operatorname{div} \varphi = \lim_j \int \nabla u_j \cdot \varphi \leq \liminf_j \int_O |\nabla u_j|.$$

Taking the supremum proves the local lower semicontinuity in (A.6). This argument requires only local integrability of u .

Finally, if $\mu(K) = 0$ for all compact Lebesgue-null K , inner regularity implies $\mu(E) = 0$ for every Borel null set E . Every Lebesgue-null set is contained in a Borel null set, so μ is absolutely continuous with respect to Lebesgue measure. Write $\mu = h \, dx$ by the positive Radon–Nikodym theorem, where $h \geq 0$ and $\int h = \mu(\mathbb{R}^2)$. Set $G = h\sigma$, with value zero where $h = 0$. Then $H = G \, dx$ and $|G| = h$ almost everywhere. Thus $\|G\|_1 = |H|(\mathbb{R}^2)$, and the defining distributional identity for H is precisely the weak-gradient identity for G . The needed inner regularity is, for example, [15, Chapter 1, Lemma 5.7]. $\square$

In Section 6, this lemma first yields a finite derivative measure from the expanding bands. The signed cutoff argument then proves that it vanishes on compact null sets; the lemma’s final assertion identifies the resulting weak gradient.

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